Simulated maize trial

11 November 2026

Contents

Summary

117 participants registered and 75 gave valid answers for ‘Overall Preference’, the main trait of this report.

8 items were tested. Each participant tested 3 of them.

Taking all answers together, the items with the best performance for ‘Overall Preference’ were Hybrid 03, Hybrid 06 and Hybrid 08.

1 of the items tested outperformed the check Hybrid 04 with statistical confidence ($p < 0.05$) for the main trait, namely Hybrid 03.

2 trait(s) were left out because they had too few valid answers, namely Flood tolerance [Vegetative] and Flood tolerance [Reproductive].

In their open answers, 84 participants talked most about early maturity and cob size. The most praised was early maturity, 71 % of its 45 mentions were positive. The main complaint was fall armyworm damage, 77 % of its 44 mentions were negative.

“ready before the rains stopped with Hybrid 01”

Results by group of participants are in the section ‘Differences between groups’.

Introduction

This report presents results from the Simulated maize trial trial implemented using the tricot approach (triadic comparison of technology options), a participatory and data-driven method for testing agricultural technologies directly in their use environments (*1*, *2*). Tricot is a citizen science approach that actively engages farmers and other users as co-researchers, enabling the generation of robust experimental evidence at scale while preserving the diversity of real-world farming conditions.

Instead of relying on a small number of tightly controlled researcher-managed trials, tricot distributes experimentation across many farms. Each participant evaluates only three randomly assigned options (for example, crop varieties, fertilizer types, or management practices) under their usual conditions. These small, simple trials are easy to implement, require minimal training, and can be conducted without continuous supervision. When aggregated across many participants and locations, the resulting data provide powerful insights into relative performance, farmer preferences, and context-specific suitability.

A key strength of the tricot approach is its focus on external validity. By allowing farmers to test technologies using their own practices, soils, and climatic conditions, the results better reflect how technologies perform in real-life settings. Farmer-reported rankings are combined with environmental, management, and socio-economic data, allowing analyses that capture genotype-by-environment interactions, preference heterogeneity, and trade-offs that are often missed in conventional trials.

Tricot is supported by a streamlined digital workflow through the ClimMob platform (*3*), which facilitates trial design, randomization, data collection, monitoring, and reporting. Statistical analysis is based on the Plackett-Luce model for ranking data (*4*, *5*), enabling valid inference from incomplete and overlapping comparisons without loss of information. Together, these methodological and digital components make tricot a scalable, cost-effective, and scientifically rigorous approach for on-farm testing, product evaluation, and decision support.

This report builds on these principles to synthesize results from the current trial cycle and to provide actionable insights for researchers, development practitioners, breeders, and other stakeholders.

What you should expect from this report

This report gives a transparent and reproducible overview of the tricot trial, from experimental design to interpretation of results. It includes the following parts.

- Trial monitoring and data quality. Participation at each data collection moment, the answers that could be used and who collected the data.
- Analytical approach. The statistical methods, mainly the Plackett-Luce model, and how environmental, management or socio-economic variables are included.
- Results and interpretation. The relative performance of the tested options, its uncertainty and patterns across environments or groups of participants.
- Measured traits. When the trial measured traits such as yield, the means of each option adjusted for the farm and the heritability.
- Evidence for variety release. A summary of each option against the check that can support a release application.

This report is produced automatically, so it cannot capture every factor that influences the performance of a technology in its target environment. For an in-depth analysis, to combine data from several projects or to prepare a peer-reviewed publication with your tricot data, send a request to the ClimMob team climmob@mrbotcr.com.

Trial monitoring and data quality

This section helps you follow the trial while it runs. It shows how many participants answered at each data collection moment, which answers can be used in the analysis and who collected the data. Use it to plan visits and reminders before the next data collection moment.

117 participants registered. Of these, 82 (70.1%) answered the last data collection moment, ‘Agronomic performance’. Figure 1 shows the answers received at each moment. The table behind it is in the appendix.

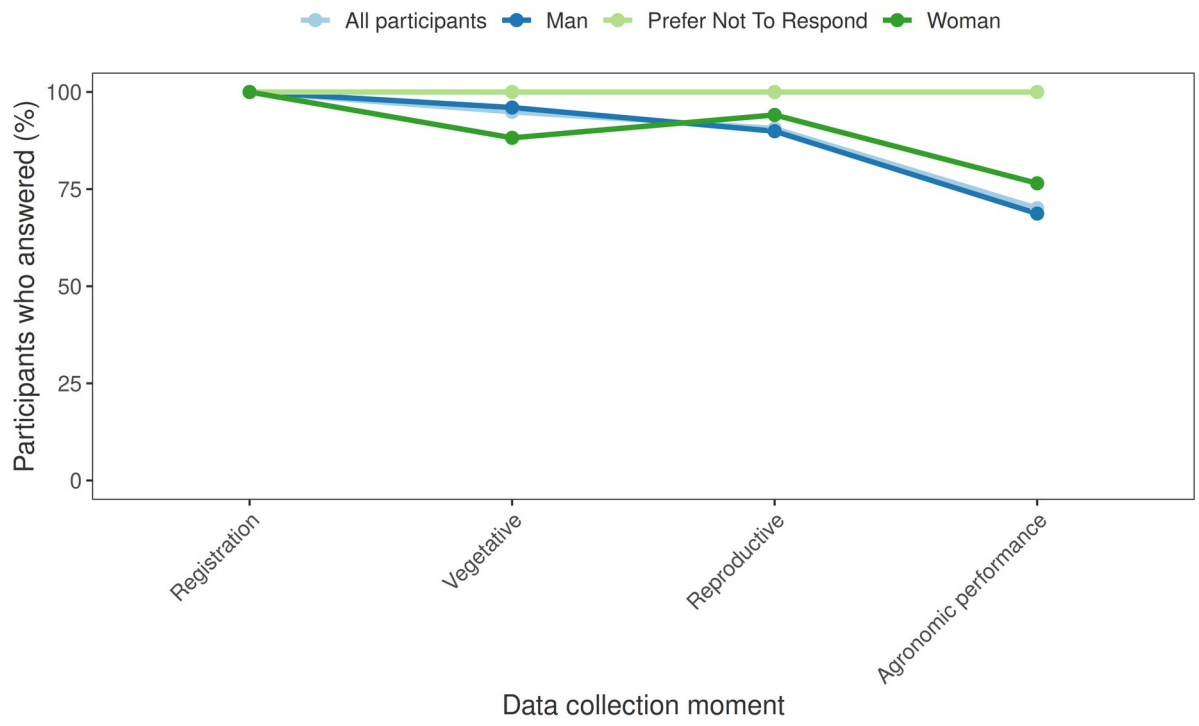


Figure 1. Percentage of registered participants who answered each data collection moment.

Table 1 shows the answers received for each trait. Only valid answers enter the analysis. An answer is not valid when the trait was not observed, when the options were tied, when the same option was given as best and worst, or when the answer was not one of the options. A trait is analysed when it has more than five valid answers and every item was tested at least twice.

Table 1. Answers received for each trait, by type.

Trait	Data collection moment	Valid	Not answered	Not observed	Tie	Same best and worst	Invalid
Germination	Vegetative	101	10	5	4	1	0
Diseases resistance	Vegetative	61	10	48	0	2	0

Trait	Data collection moment	Valid	Not answered	Not observed	Tie	Same best and worst	Invalid
Drought tolerance	Vegetative	62	10	49	0	0	0
Flood tolerance	Vegetative	5	10	106	0	0	0
Fall armyworm resistance	Vegetative	82	10	26	2	1	0
Diseases resistance	Reproductive	55	15	51	0	0	0
Drought tolerance	Reproductive	46	15	58	1	1	0
Flood tolerance	Reproductive	9	15	96	1	0	0
Fall armyworm resistance	Reproductive	81	15	24	1	0	0
Maturity	Agronomic performance	78	39	1	2	1	0
Yield	Agronomic performance	79	39	1	2	0	0
Grain size	Agronomic performance	77	39	2	1	2	0
Grain colour	Agronomic performance	78	39	2	2	0	0
Taste	Agronomic performance	77	39	3	1	1	0
Marketability	Agronomic performance	72	39	5	3	2	0
Overall preference	Agronomic performance	75	39	3	2	2	0

These traits were left out of the analysis because they had too few valid answers, namely Flood tolerance [Vegetative] and Flood tolerance [Reproductive].

The number of forms submitted by each person who collected data is listed in the appendix.

Location of the trial plots

Figure 2 shows where the trial plots were located.

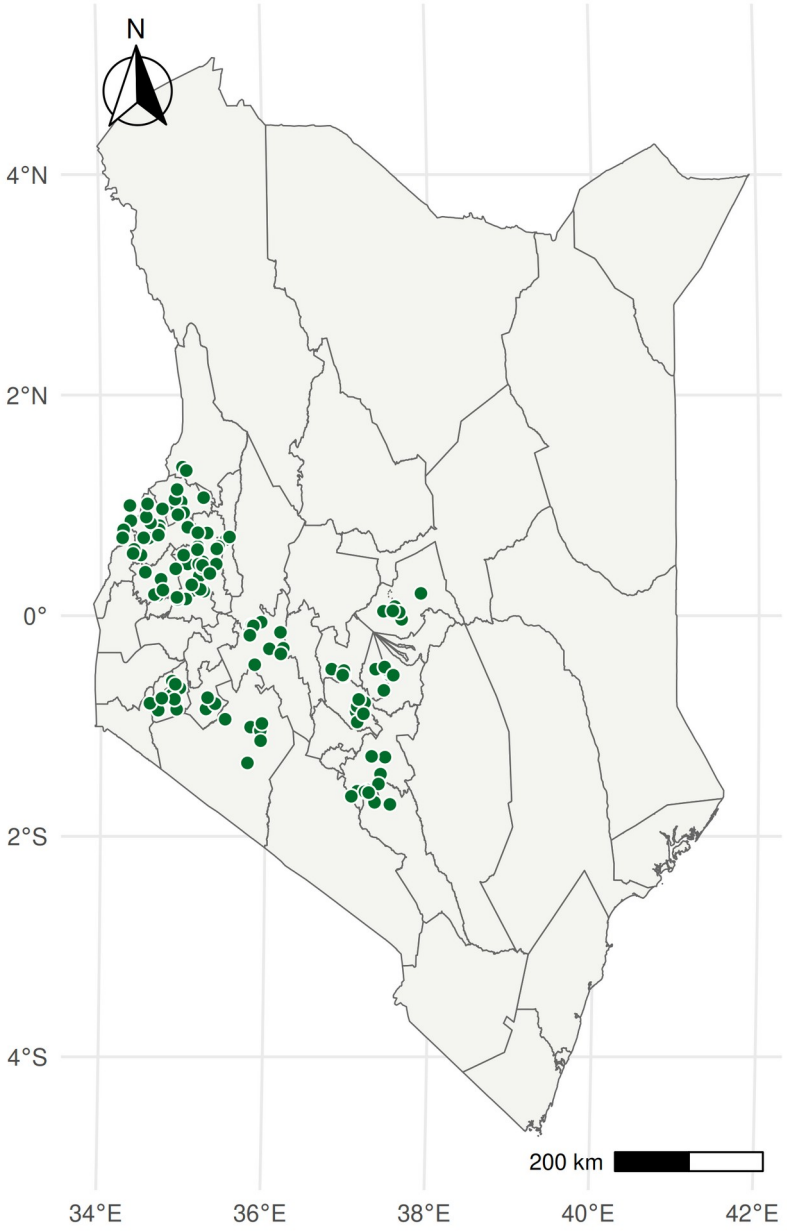


Figure 2. Location of the trial plots. Administrative boundaries are from the Database of Global Administrative Areas (GADM).

Who answered the latest data collection

Figure 3 shows the trial plots by whether the participants answered the latest data collection, «Agronomic performance». 82 of the 121 participants with GPS coordinates answered. Use the map to plan visits and reminders.

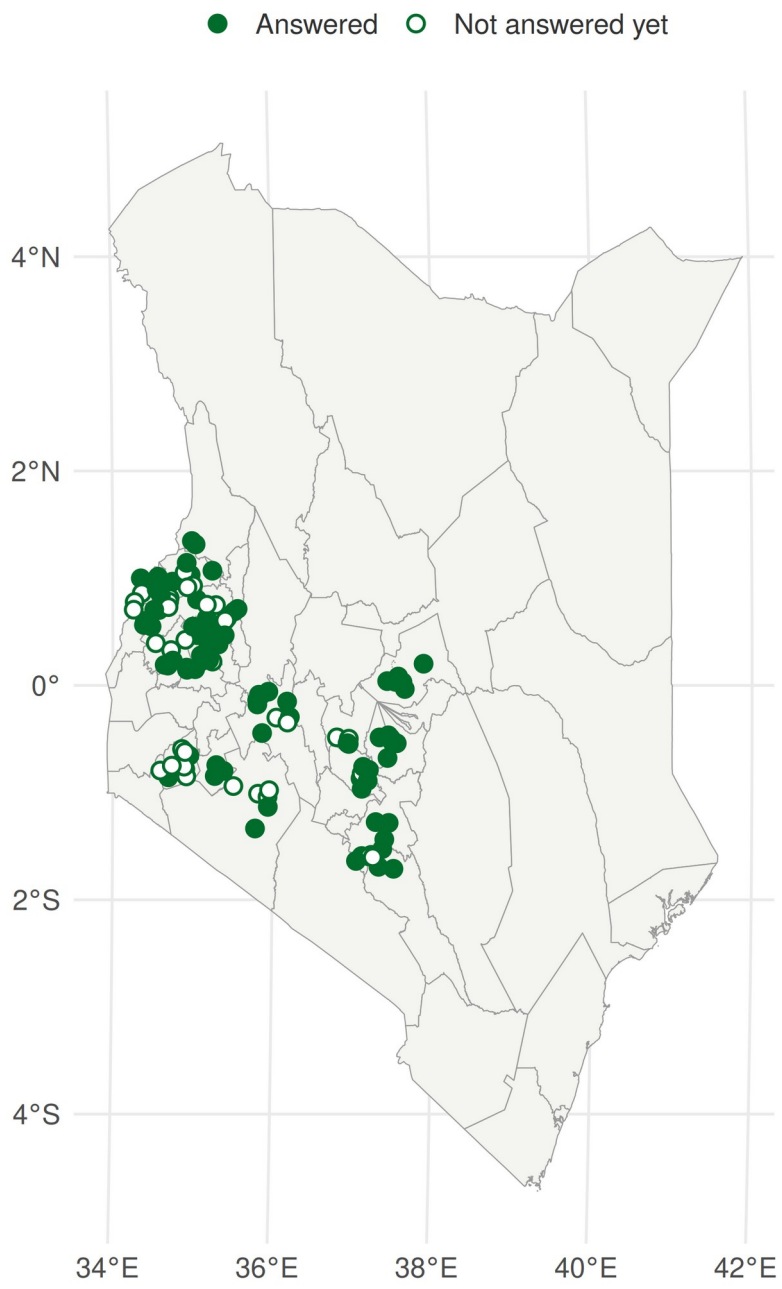


Figure 3. Trial plots by answer at «Agronomic performance». Filled points answered, open points have not answered yet.

Climate during the season

Figure 4 shows the season at the trial plots in the map, for 85 days from the planting day (from 2022-09-03 to 2022-12-08), with the data collection moments marked. The window follows the data collection moments set for this project. Across the plots, the median total rainfall was 354 mm (from 139 to 472 mm in the 10th to 90th percentile), the longest dry spell lasted 8 days, day temperature reached 25.9 °C and nights went down to 11.8 °C (medians over the window). Data come from NASA POWER through the R package nasapower (*6*) and the indices were

computed with the R package climatrends (*7*). The same indices are tested as covariates in the Plackett-Luce trees. The weekly indices and the climate envelope of the trial are in the appendix.

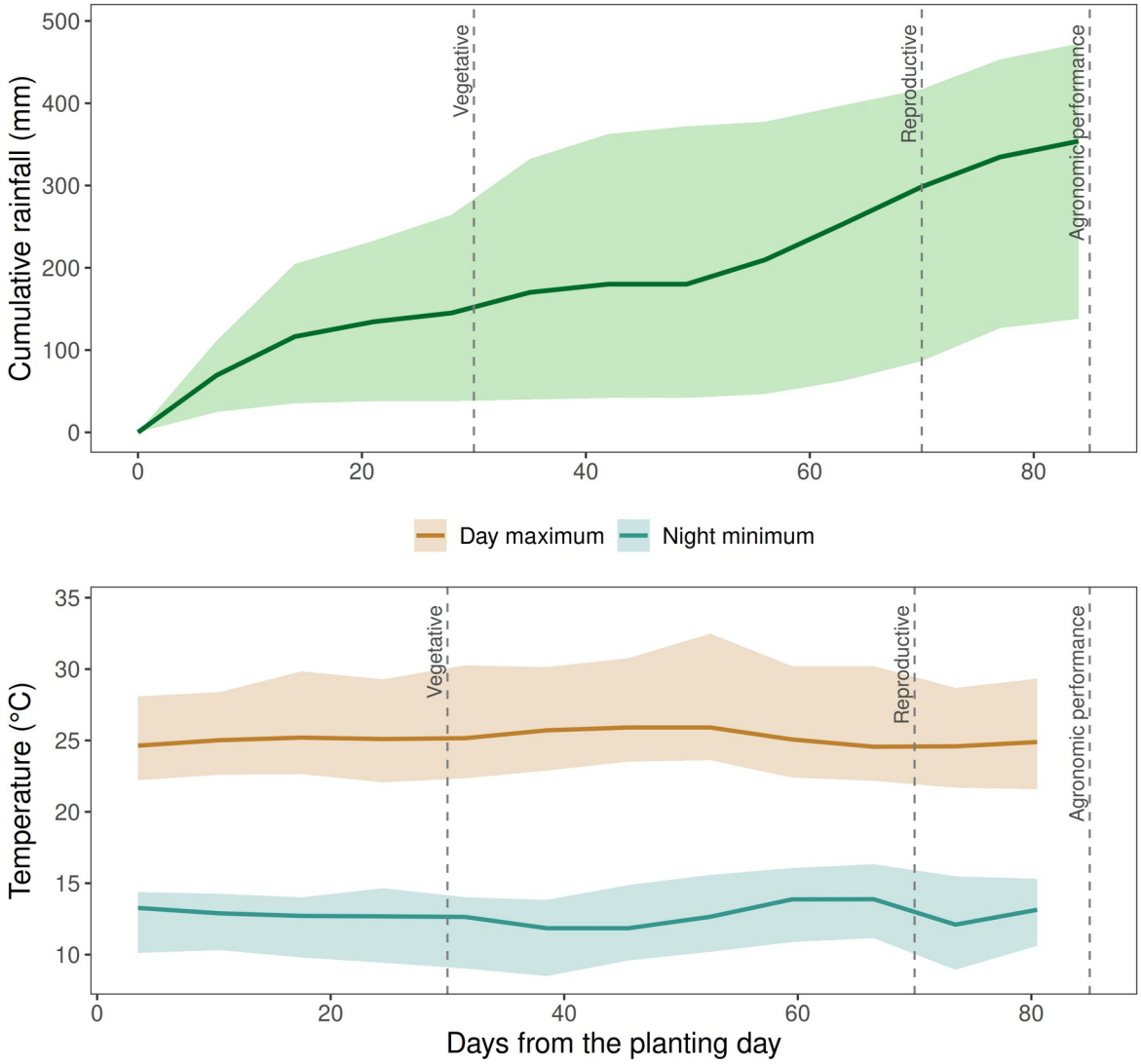


Figure 4. The season at the trial plots, from 2022-09-03 to 2022-12-08. Cumulative rainfall (top) and maximum day and minimum night temperature (bottom), as the median across plots with the band from the 10th to the 90th percentile. Dashed lines mark the data collection moments, in days from the planting day.

How many answers are needed?

Each answer makes the results more reliable. To show how much, the trial was simulated with the preferences estimated for «Overall Preference» taken as the truth. Each simulated participant tests 3 items drawn at random, and the model is refitted 50 times for each number of participants. Figure 5 shows how well the simulated trials recover the order of the items (Kendall tau) and how often they find the same best item. With the 75 answers available now, the chance of finding the best item is 90 %.

If all 117 registered participants answered, the chance would be 94 %.

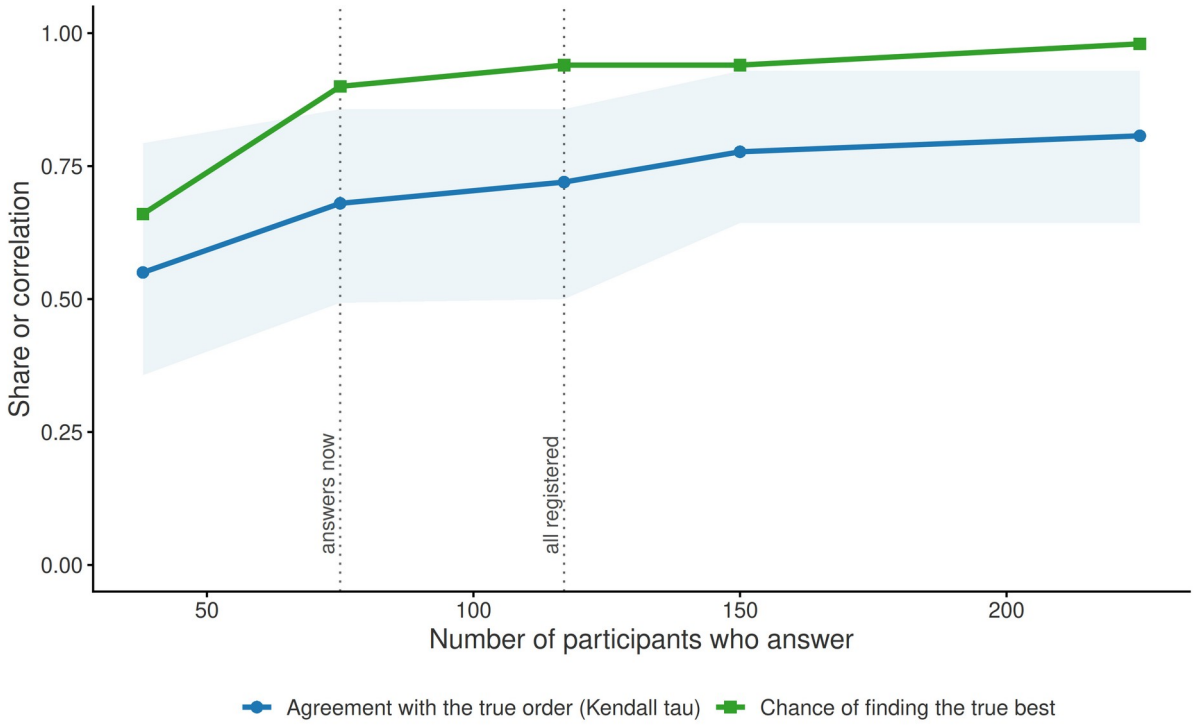


Figure 5. Expected accuracy of the ranking for «Overall Preference» by the number of participants who answer. The band shows the 10th to 90th percentiles of Kendall tau.

Methods

ClimMob applies the Plackett-Luce (PL) model, independently proposed by Luce (1959) (*5*) and Plackett (1975) (*4*), to analyse ranking data. It is implemented in R by the package 'PlackettLuce' (*8*). The PL model estimates the probability that one technology outperforms the others following Luce's Choice Axiom (*5*), which states that the probability that one item outperforms another is independent of the presence or absence of any other item in the set.

Here is a positive score for item , and reads as ' is preferred over '. The PL model estimates these scores, called *worth*, which give the probability that each item outperforms all other items in the set. We report *log-worth* centred on the check Hybrid 04 and Hybrid 05. Each tricot ranking of three technologies has the following probability under the PL model.

Covariates, if any, are linked to the rankings with model-based recursive partitioning (*9*), which builds Plackett-Luce trees. The trees find groups of participants for which the performance of the technologies differs. When coordinates and dates are available, rainfall and temperature indices from NASA POWER are added as covariates (*6*, *7*). Covariates are chosen by forward selection, adding at each step the covariate that most improves the model according to the Akaike Information Criterion (AIC) (*10*). The AIC adds the degrees of freedom () to the maximised log-likelihood ().

Based on this approach, the following model was applied to the data.

```
G ~ Age + Education
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Not all covariates in this model appear in the tree, since the tree is limited to a few nodes. The covariates shown are the most important for this data set. This report uses 'Overall Preference' as the reference trait for the analysis.

The correlation between 'Overall Preference' and the other traits is estimated with the Kendall tau coefficient (*11*), an equivalent of the Pearson correlation for ranking data. It ranges from -1 to 1, where 1 means full agreement and -1 full disagreement. It can show what drives the choices of participants and help to prioritise traits in the next stage of testing.

This report is generated in R (*12*) with the packages 'knitr' (*13*) and 'rmarkdown' (*14*). Data handling and analysis use 'ClimMobTools' (*15*), 'gosset' (*16*), 'PlackettLuce' (*8*), 'partykit' (*17*),

‘psychotools’ (*18*), ‘qvcalc’ (*19*), ‘lme4’ (*20*), ‘gtools’ (*21*) and ‘jsonlite’ (*22*). Charts use ‘ggplot2’ (*23*), ‘ggparty’ (*24*), ‘igraph’ (*25*), ‘multcompView’ (*26*) and ‘patchwork’ (*27*). The tricot approach is introduced by van Etten et al. (2019) (*2*). To cite ClimMob, use van Etten et al. (2020) (*28*). The workflow that produced this report is described by de Sousa et al. (2022) (*29*).

Trial design

121 participants took part in this trial. Each assessed 3 different items and ranked them for 14 trait(s). The traits analysed and the number of valid answers for each are listed in the appendix.

Figure 6 shows the experimental design as a network. Two items are connected when they were tested together in at least one package of 3. The network uses the rankings of 'Overall Preference'.

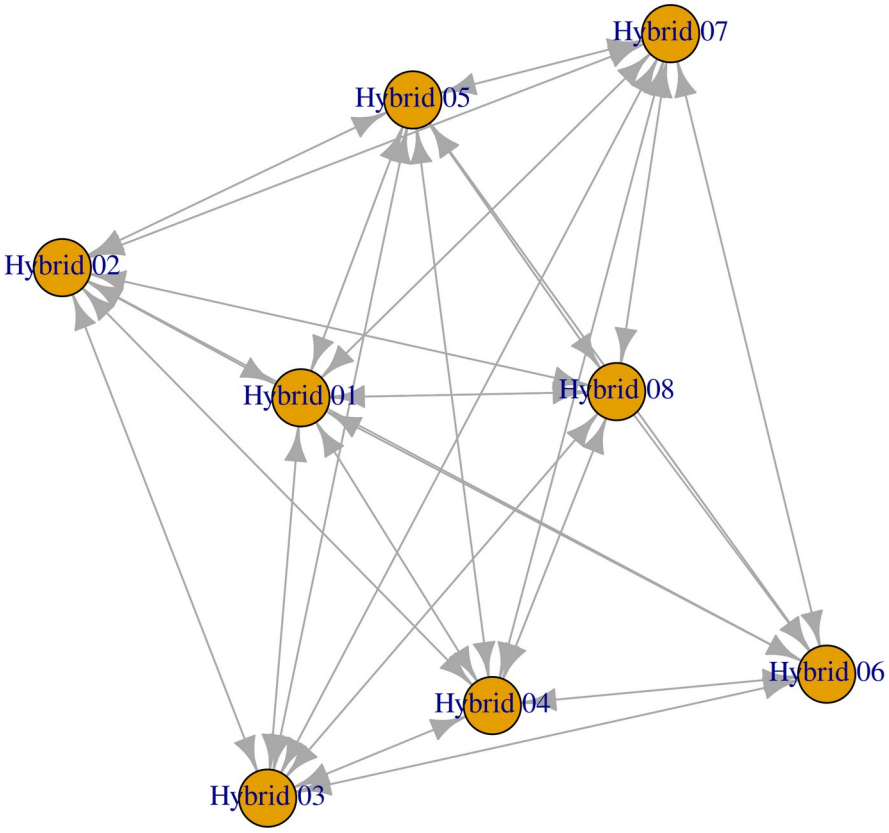


Figure 6. Network of the items tested in this trial. Arrows show direct wins and losses between pairs of items that were tested together.

Table 2 lists the 8 items tested, with the number and percentage of participants who tested each. Names longer than 12 characters are abbreviated in the charts.

Table 2. Frequency of the items tested.

Item	Abbreviation	Frequency	Relative frequency
Hybrid 01	Hybrid 01	45	37.2%
Hybrid 02	Hybrid 02	38	31.4%
Hybrid 03	Hybrid 03	40	33.1%
Hybrid 04	Hybrid 04	49	40.5%
Hybrid 05	Hybrid 05	59	48.8%
Hybrid 06	Hybrid 06	40	33.1%
Hybrid 07	Hybrid 07	40	33.1%
Hybrid 08	Hybrid 08	52	43%

Performance of the items

Table 3 tests, for each trait, whether at least one item performs differently from the others (likelihood ratio test against a model where all have the same worth). A p-value below 0.05 means that the differences in performance are unlikely to be due to chance.

Table 3. Test of differences between items, by trait.

Trait	logLik	DF	Statistic	p-value	
Germination [Vegetative]	-166.916	296	28.104	< 0.001	***
Diseases resistance [Vegetative]	-89.658	176	39.278	< 0.001	***
Drought tolerance [Vegetative]	-99.911	179	22.356	0.0022	**
Fall armyworm resistance [Vegetative]	-138.628	239	16.592	0.0202	*
Diseases resistance [Reproductive]	-94.560	158	7.974	0.335	
Drought tolerance [Reproductive]	-73.060	131	18.722	0.0091	**
Fall armyworm resistance [Reproductive]	-136.195	236	17.876	0.0125	*
Maturity [Agronomic performance]	-129.845	227	19.824	0.006	**
Yield [Agronomic performance]	-129.411	230	24.276	0.001	***
Grain size [Agronomic performance]	-127.166	224	21.598	0.003	**
Grain colour [Agronomic performance]	-127.196	227	25.122	< 0.001	***
Taste [Agronomic performance]	-124.991	224	25.948	< 0.001	***
Marketability [Agronomic performance]	-122.166	209	13.682	0.0571	.
Overall preference [Agronomic performance]	-125.972	218	16.820	0.0186	*

Table 4 gives the favourability of each item for the main trait and, when it was ranked, yield. ‘Top’ and ‘Bottom’ are the probabilities (%) that the item was ranked first or last in the sets where it was tested, and the favourability score is their difference. Positive values mean that the item was more often a winner than a loser. The same table for every trait is in the appendix. Breeding teams use it in variety release applications.

Table 4. Favourability of the items for the main trait and yield.

Trait	Option	N	Top (%)	Bottom (%)	Favourability
Yield [Agronomic Performance]	Hybrid 03	56	50.0	16.1	33.9
	Hybrid 06	52	49.7	15.9	33.7

Trait	Option	N	Top (%)	Bottom (%)	Favourability
	Hybrid 08	66	37.1	26.5	10.6
	Hybrid 02	40	36.5	28.5	8.1
	Hybrid 05	76	29.6	35.4	-5.8
	Hybrid 01	62	28.6	34.8	-6.1
	Hybrid 04	70	26.3	41.1	-14.8
	Hybrid 07	52	12.4	66.6	-54.2
Overall preference [Agronomic Performance]	Hybrid 03	54	57.3	11.3	46.0
	Hybrid 06	42	44.6	20.1	24.5
	Hybrid 08	62	36.6	27.7	8.8
	Hybrid 02	42	33.8	30.5	3.3
	Hybrid 05	70	28.6	37.5	-8.9
	Hybrid 01	58	26.5	39.2	-12.7
	Hybrid 07	54	23.0	46.0	-23.0
	Hybrid 04	68	23.1	46.5	-23.5

Figure 7 shows the Kendall tau between ‘Overall Preference’ and the other traits. The strongest correlation was with ‘Drought tolerance [Vegetative]’ (tau = 0.14) and the weakest with ‘Fall armyworm resistance [Vegetative]’ (tau = -0.05). A table with the significance tests is in the Appendix.

The bars show 95 % intervals from 200 bootstrap resamples of participants. An interval that does not cross zero indicates a clear association.

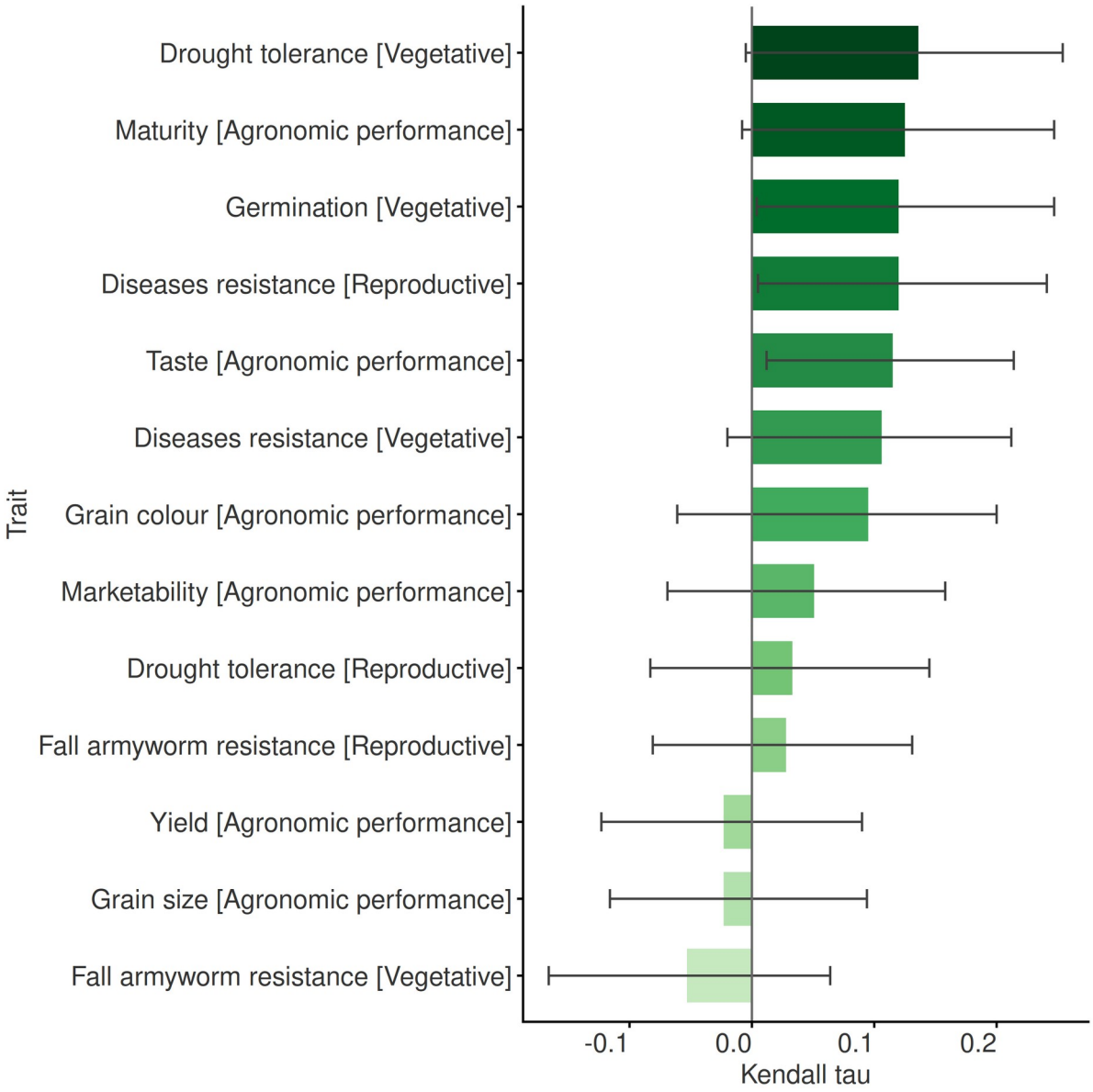


Figure 7. Kendall correlation between 'Overall Preference' and the other traits.

Figure 8 shows the log-worth of each item for every trait, relative to the check Hybrid 04 and Hybrid 05. Positive values show items that performed better than the check for that trait, and negative values those that performed worse.

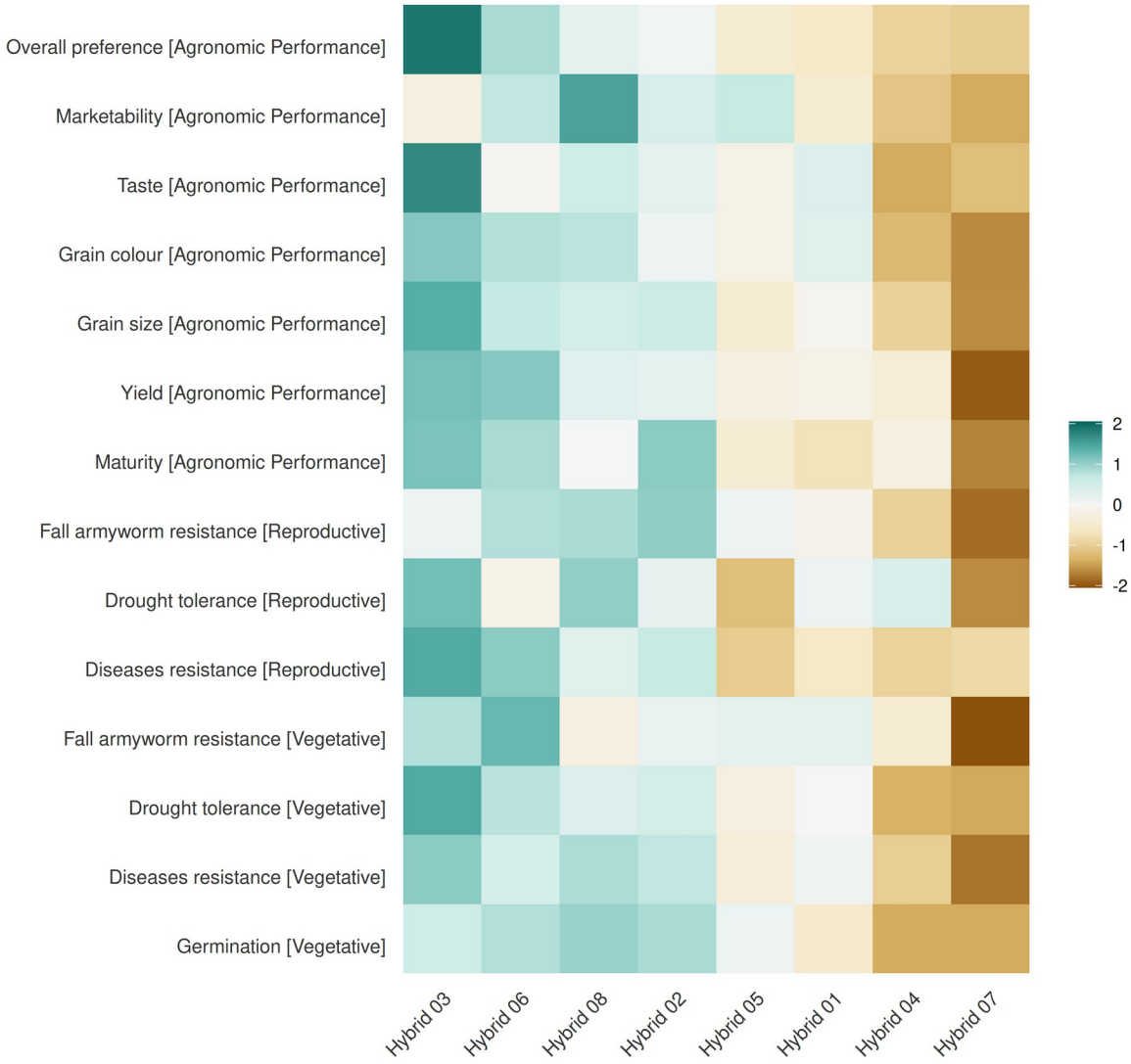


Figure 8. Item performance by trait, as log-worth relative to the check.

Figure 9 shows the reliability of each item, the probability that it outperforms the check Hybrid 04 (*30*). Values above 0 mean that the item is more likely to beat the check than to lose to it. Results for other checks are in the extra outputs.

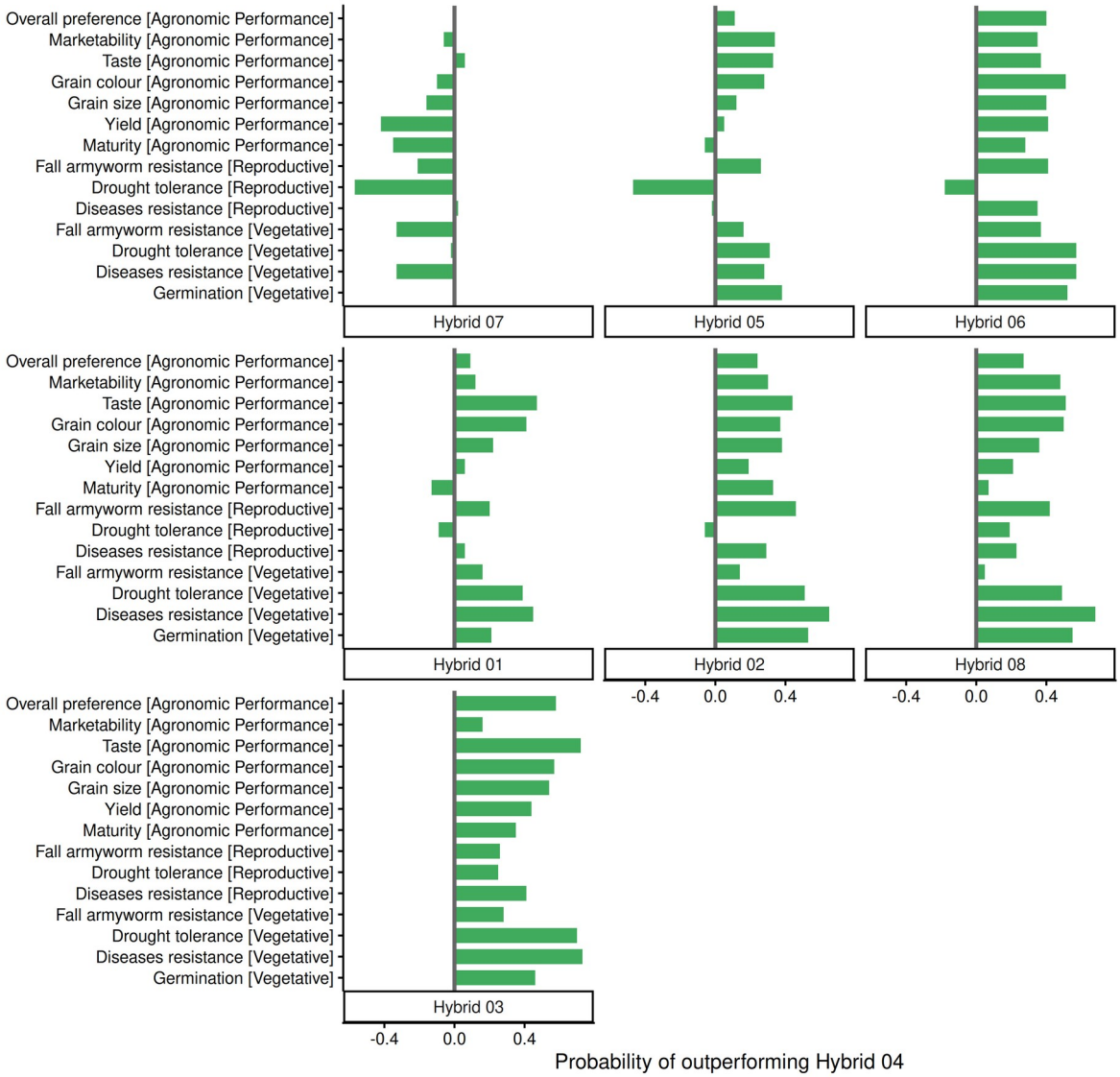


Figure 9. Reliability of the items against the check Hybrid 04.

Figure 10 shows the log-worth of each item for 'Overall Preference'. items that share a letter are not significantly different (p = 0.05). Bars show 50% confidence intervals from quasi standard errors.

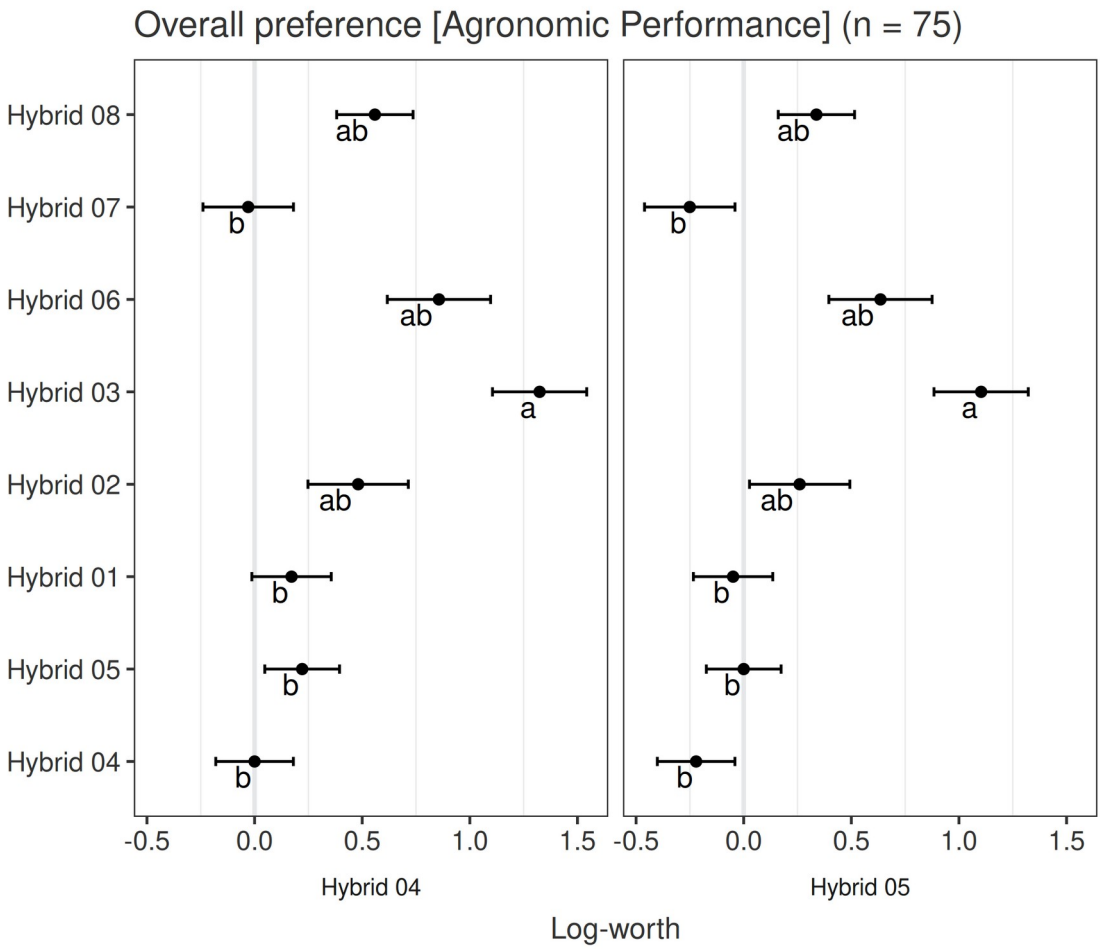


Figure 10. Log-worth of the items for 'Overall Preference'.

Why they preferred it

The open answers read by Sikia Analytics say why participants ranked the Overall Preference as they did. Table 5 lists, for every option, the topics with mostly positive mentions and those with mostly negative mentions, with the number of mentions in brackets.

Hybrid 03, the item ranked first, was praised for storage weevils (5), drought tolerance (4). The complaints were about early maturity (13), lodging (4), fall armyworm damage (3).

Table 5. What participants said about each of the items, from the open answers read by Sikia Analytics. Options in the order of the ranking for Overall Preference.

Option	Praised for	Complaints about	Mentions
Hybrid 03	storage weevils (5), drought tolerance (4)	early maturity (13), lodging (4), fall armyworm damage (3)	37
Hybrid 06	fall armyworm damage (9), cob size (5), lodging (3)	drought tolerance (3), storage weevils (2)	42
Hybrid 02	cob size (7), lodging (4)	early maturity (12), drought tolerance (6), fall armyworm damage (6)	43
Hybrid 05	early maturity (7), lodging (6), drought tolerance (3)	taste (8), fall armyworm damage (6), seed cost (6)	43

Option	Praised for	Complaints about	Mentions
Hybrid 01	early maturity (11), drought tolerance (5), storage weevils (4)	lodging (7), taste (3)	42
Hybrid 04	fall armyworm damage (9), taste (9), seed cost (7)	cob size (6), storage weevils (2)	47

Differences between groups

The results were requested by group of participants. A comparison between groups is only reliable when every group has enough answers, so each group needs at least 20 valid answers for ‘Overall Preference’ to be analysed. Table 6 shows the groups found in the data.

Table 6. Groups of participants and valid answers for the main trait.

Group	Valid answers	Share	Status
Man	63	84%	analysed
Woman	12	16%	too few answers

Only 12 participants in the group ‘Woman’ gave valid answers (16% of those with a group). This is too few for a separate result. The overall results therefore mostly reflect the other groups. To compare groups in future trials, aim for at least 20 participants in every group.

Fewer than two groups have enough answers, so the groups are not compared. The results in this report mostly reflect the views of the largest group, ‘Man’ (84% of the participants with a group).

Effect of covariates on the performance of the items

Table 7 presents the groups (nodes) found by the Plackett-Luce tree with the covariates kept by the forward selection.

Table 7. Groups (nodes) identified by the Plackett-Luce tree.

Node	N	Superior performance	Underperformance	Variable and threshold
2	28	Hybrid 02, Hybrid 06, Hybrid 08	Hybrid 05	Age <= 38
4	20	Hybrid 02, Hybrid 03, Hybrid 05	Hybrid 04, Hybrid 07, Hybrid 06	Age > 38 & Education = Illiterate, Primary School
5	27	Hybrid 03, Hybrid 06, Hybrid 07	Hybrid 02, Hybrid 08, Hybrid 04	Age > 38 & Education = High School, Other, Secondary School, University Degree, Vocational Training

Figure 11 shows the Plackett-Luce tree.

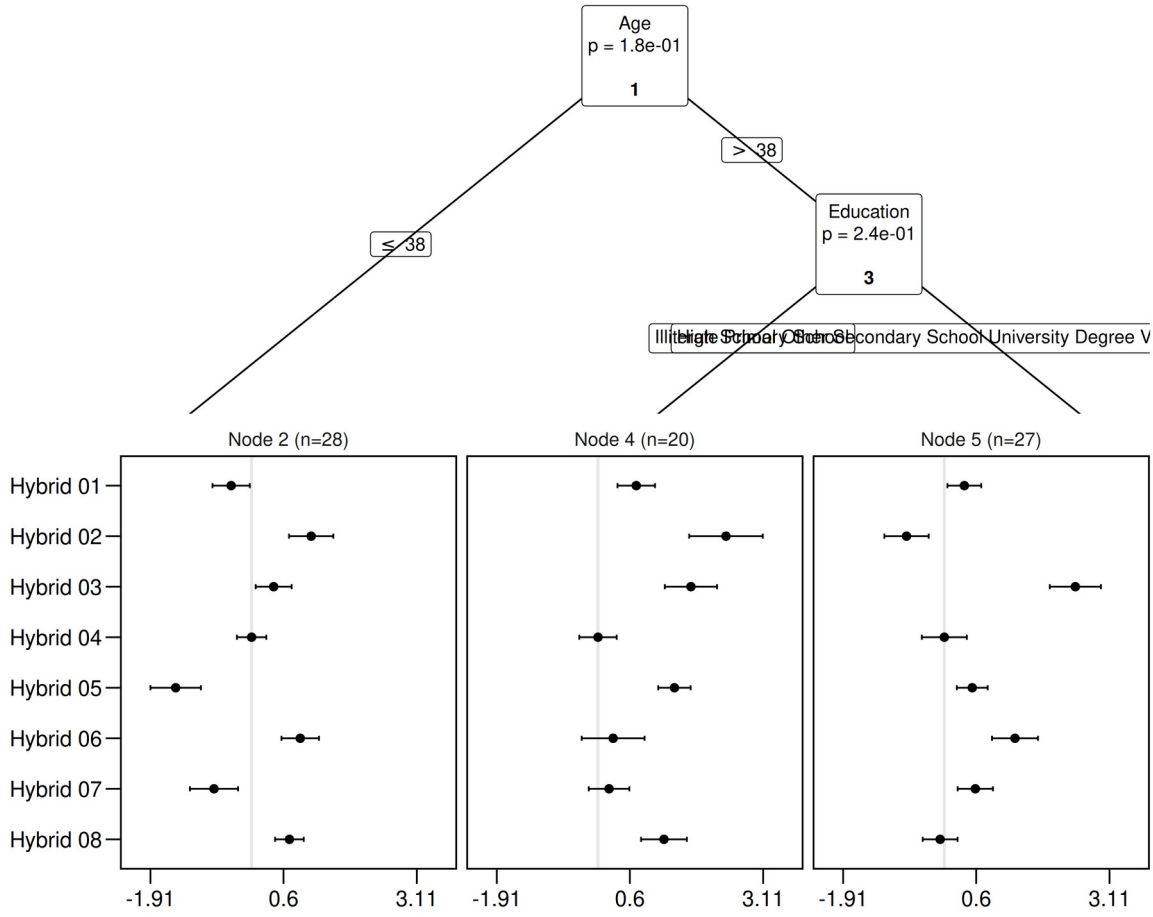


Figure 11. Plackett-Luce tree of the items. The horizontal axis of each panel is the log-worth. Error bars show quasi standard errors.

Figure 12 shows where the groups of the tree are and how the participants spread around each threshold.

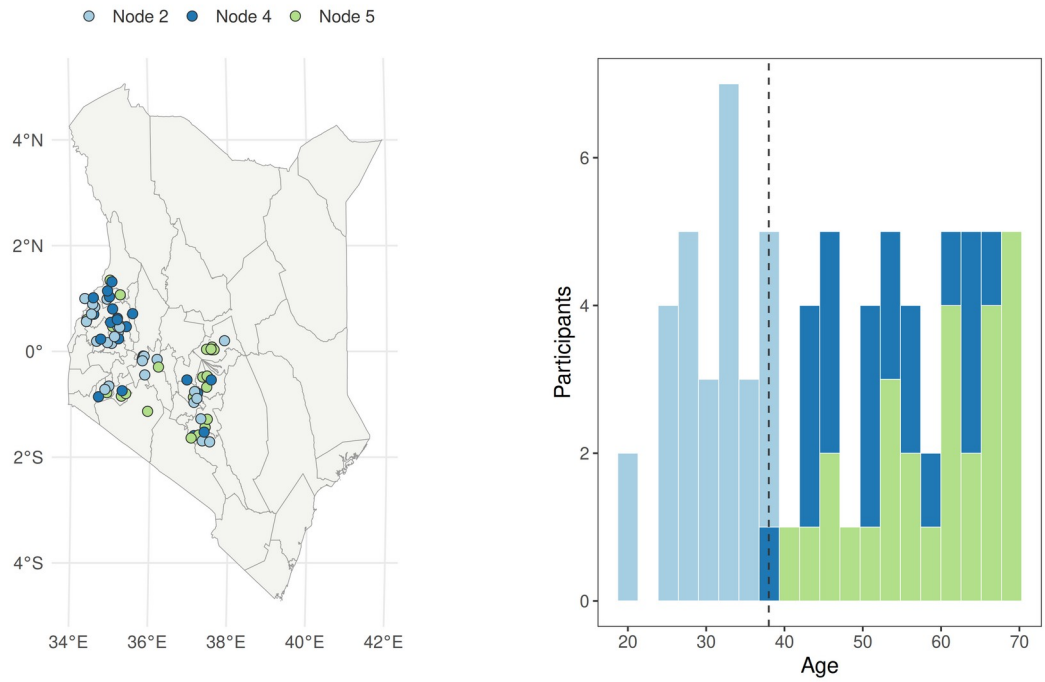


Figure 12. Trial plots coloured by the node of the Plackett-Luce tree they fall in (left), and the distribution of each numeric split variable across participants with the threshold of the split as a dashed line (right).

Table 8 gives the worth and regret of the items across the nodes of the tree (*31*, *32*). Worst regret is the loss under the worst possible outcome. Choosing items with a low worst regret is a cautious way to manage risk across all the groups in the tree.

Table 8. Probability of winning (worth) and regret of each item. High worth and low regret indicate a good choice.

Option	Worth	Worst regret	Regret
Hybrid 03	0.282	0.051	0.021
Hybrid 06	0.142	0.107	0.106
Hybrid 02	0.215	0.150	0.113
Hybrid 08	0.110	0.145	0.138
Hybrid 05	0.079	0.134	0.140
Hybrid 01	0.064	0.137	0.154
Hybrid 07	0.055	0.133	0.159
Hybrid 04	0.054	0.143	0.169

Measured traits

This section analyses traits measured on each plot, such as yield. Zeros are treated as missing plots. Values outside 1.5 times the interquartile range are flagged as possible outliers, listed in the Appendix and left out of the model. The mean of each item is estimated with a linear mixed model with the farm of each participant as a random effect (*20*), so that differences between farms do not bias the comparison. Broad-sense heritability is estimated with the ad hoc method of Holland et al. (2003) (*33*).

Plant population - Vegetative

363 plots from 121 farms. 30 plots had no value (0 values were zero) and 1 values were flagged as possible outliers.

Table 9. Mean of each item adjusted for the farm, with 95% confidence interval and comparison with the check Hybrid 04. Items that share a letter are not significantly different (p = 0.05).

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 01	38	51.05	49.11	52.99	104.85	0.074	a
Hybrid 08	47	50.93	49.18	52.68	104.59	0.0728	a
Hybrid 05	55	49.76	48.14	51.37	102.18	0.373	ab
Hybrid 02	37	49.62	47.65	51.59	101.90	0.487	ab
Hybrid 07	37	49.62	47.65	51.59	101.90	0.488	ab
Hybrid 03	37	49.06	47.10	51.03	100.76	0.782	ab
Hybrid 04	45	48.69	46.91	50.48	100.00		ab
Hybrid 06	36	47.68	45.69	49.67	97.92	0.45	b

Broad-sense heritability is 0.28. Values close to 1 mean that most of the variation between plots comes from differences between the items, values close to 0 that it comes from the environment and measurement error.

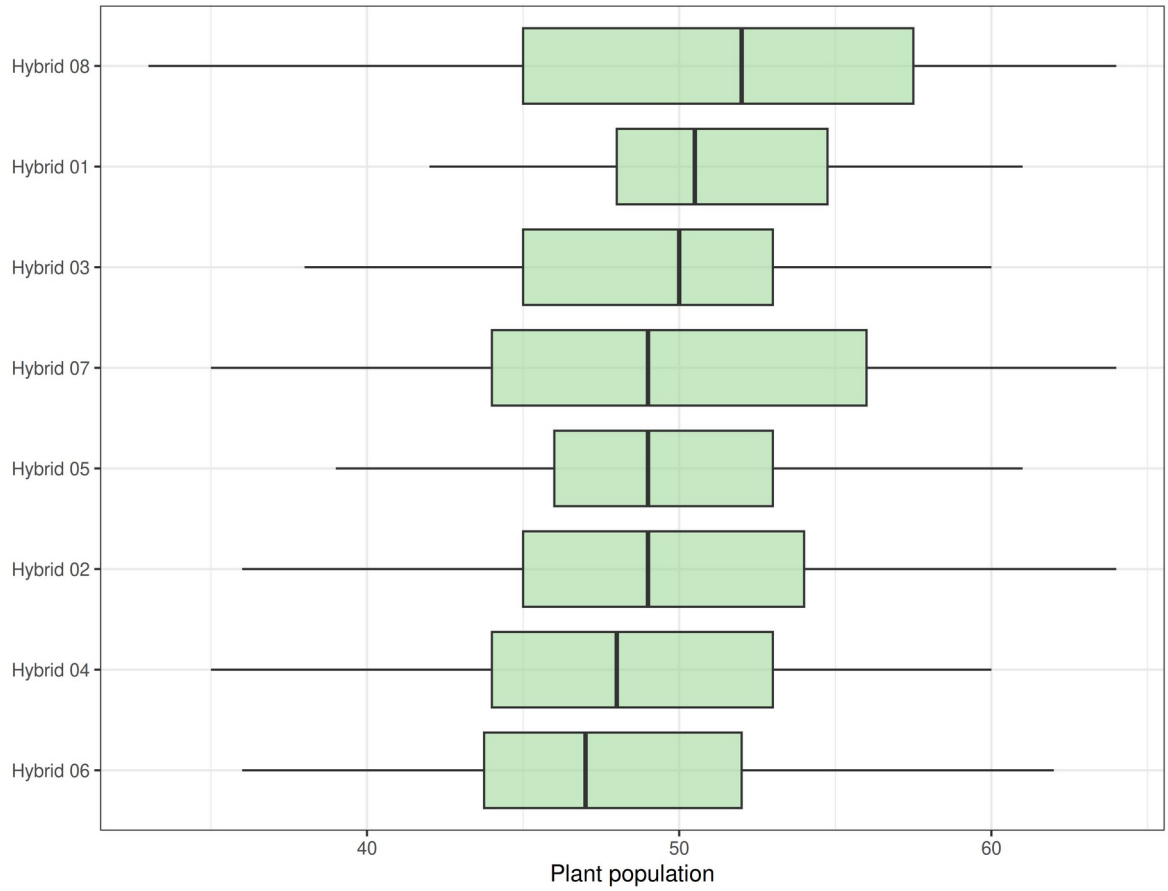


Figure 13. Distribution of Plant population - Vegetative by option, possible outliers removed.

Disease severity - Vegetative

363 plots from 121 farms. 35 plots had no value (5 values were zero) and 3 values were flagged as possible outliers.

Table 10. Mean of each item adjusted for the farm, with 95% confidence interval and comparison with the check Hybrid 04. Items that share a letter are not significantly different (p = 0.05).

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 07	35	30.94	27.75	34.14	111.23	0.153	b
Hybrid 04	44	27.82	24.97	30.67	100.00		ab
Hybrid 01	38	25.42	22.35	28.49	91.38	0.262	ac
Hybrid 05	55	23.22	20.67	25.77	83.46	0.0184	cd
Hybrid 02	35	21.97	18.77	25.17	78.98	0.00747	cde
Hybrid 06	37	19.95	16.84	23.06	71.70	< 0.001	def
Hybrid 08	47	19.45	16.69	22.21	69.91	< 0.001	ef
Hybrid 03	34	16.53	13.29	19.77	59.42	< 0.001	f

Broad-sense heritability is 0.89. Values close to 1 mean that most of the variation between plots comes from differences between the items, values close to 0 that it comes from the environment and measurement error.

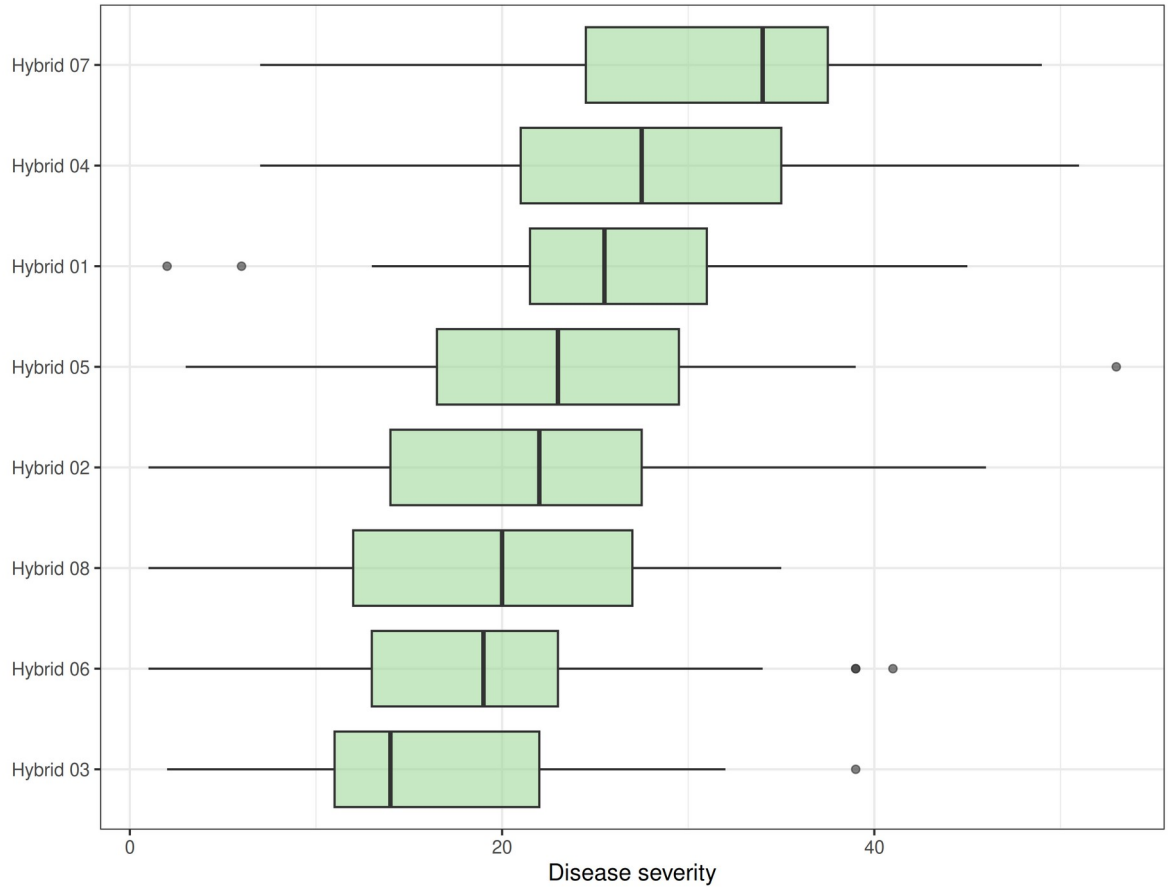


Figure 14. Distribution of Disease severity - Vegetative by option, possible outliers removed.

Plant population - Reproductive

363 plots from 121 farms. 45 plots had no value (0 values were zero) and 3 values were flagged as possible outliers.

Table 11. Mean of each item adjusted for the farm, with 95% confidence interval and comparison with the check Hybrid 04. Items that share a letter are not significantly different (p = 0.05).

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 03	37	50.47	48.48	52.45	103.94	0.174	a
Hybrid 01	39	50.25	48.32	52.19	103.50	0.221	a
Hybrid 05	50	49.68	47.97	51.39	102.32	0.389	a
Hybrid 08	45	49.40	47.60	51.20	101.74	0.529	a
Hybrid 07	37	49.36	47.37	51.34	101.66	0.568	a
Hybrid 06	35	48.75	46.71	50.78	100.40	0.893	a
Hybrid 04	39	48.55	46.62	50.48	100.00		a

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 02	33	47.59	45.49	49.69	98.02	0.508	a

Broad-sense heritability is 0. Values close to 1 mean that most of the variation between plots comes from differences between the items, values close to 0 that it comes from the environment and measurement error.

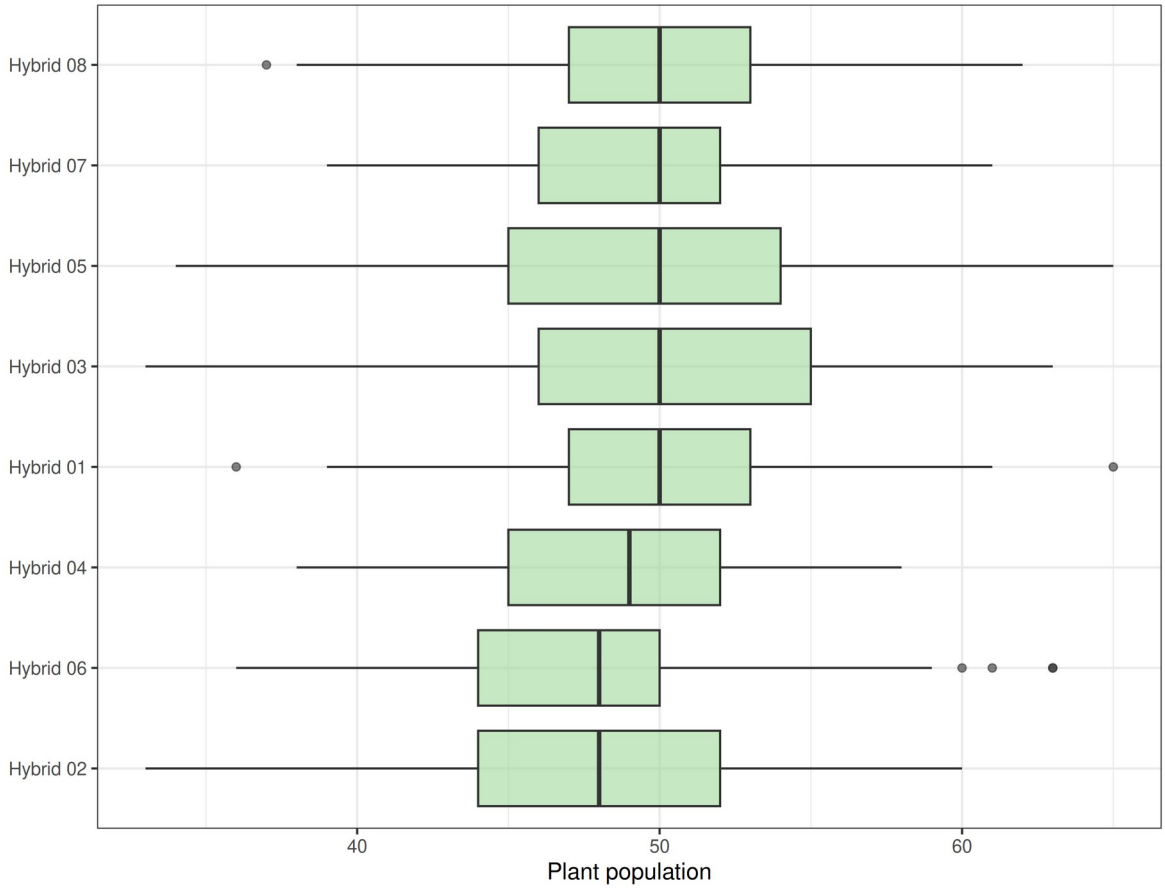


Figure 15. Distribution of Plant population - Reproductive by option, possible outliers removed.

Disease severity - Reproductive

363 plots from 121 farms. 50 plots had no value (5 values were zero) and 1 values were flagged as possible outliers.

Table 12. Mean of each item adjusted for the farm, with 95% confidence interval and comparison with the check Hybrid 04. Items that share a letter are not significantly different (p = 0.05).

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 07	36	32.89	29.71	36.07	108.33	0.261	a
Hybrid 04	39	30.36	27.30	33.42	100.00		a
Hybrid 02	32	24.00	20.62	27.38	79.05	0.00621	b
Hybrid 01	40	23.93	20.91	26.94	78.81	0.00334	b

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 05	50	23.14	20.44	25.84	76.22	< 0.001	b
Hybrid 08	44	20.66	17.78	23.54	68.05	< 0.001	b
Hybrid 06	34	19.79	16.52	23.07	65.20	< 0.001	bc
Hybrid 03	37	16.32	13.18	19.46	53.77	< 0.001	c

Broad-sense heritability is 0.92. Values close to 1 mean that most of the variation between plots comes from differences between the items, values close to 0 that it comes from the environment and measurement error.

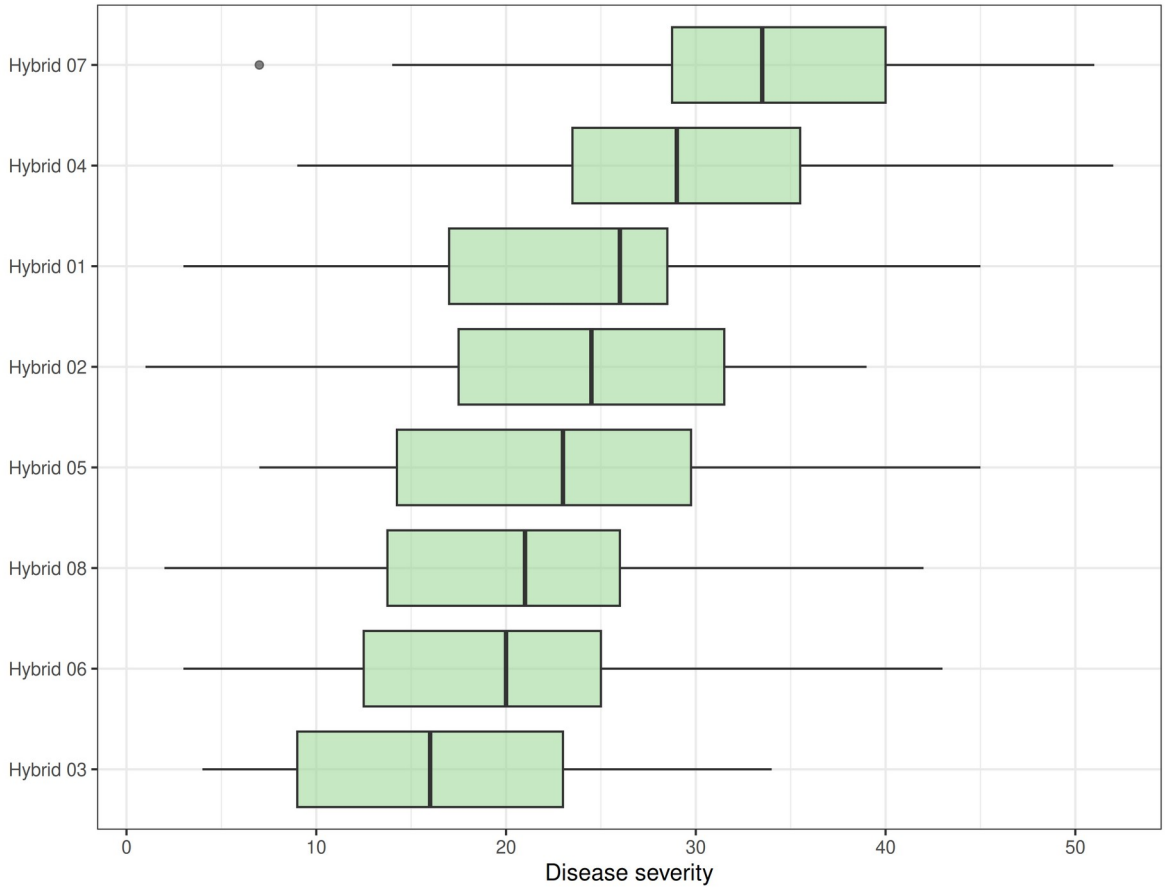


Figure 16. Distribution of Disease severity - Reproductive by option, possible outliers removed.

Number of cobs - Agronomic performance

363 plots from 121 farms. 117 plots had no value (0 values were zero) and 2 values were flagged as possible outliers.

Table 13. Mean of each item adjusted for the farm, with 95% confidence interval and comparison with the check Hybrid 04. Items that share a letter are not significantly different (p = 0.05).

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 03	30	48.11	45.23	50.99	132.79	< 0.001	d

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 06	26	43.15	40.05	46.24	119.09	< 0.001	c
Hybrid 02	22	41.56	38.19	44.92	114.70	0.0146	c
Hybrid 05	38	41.22	38.66	43.78	113.78	0.00773	c
Hybrid 08	35	40.44	37.78	43.11	111.63	0.0279	c
Hybrid 01	32	39.49	36.70	42.28	109.00	0.0964	ac
Hybrid 04	35	36.23	33.56	38.90	100.00		ab
Hybrid 07	26	33.90	30.81	37.00	93.58	0.263	b

Broad-sense heritability is 0.88. Values close to 1 mean that most of the variation between plots comes from differences between the items, values close to 0 that it comes from the environment and measurement error.

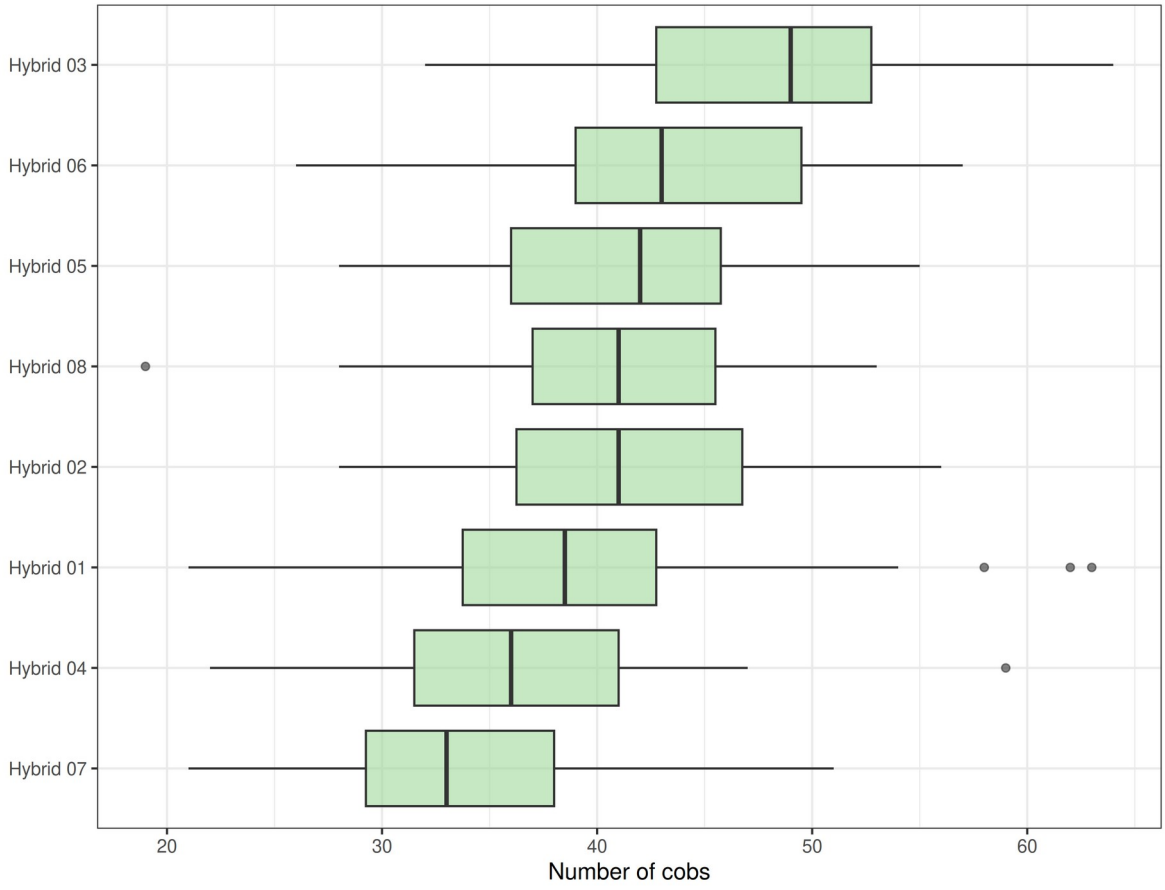


Figure 17. Distribution of Number of cobs - Agronomic performance by option, possible outliers removed.

Yield (quantitative) - Agronomic performance

363 plots from 121 farms. 145 plots had no value (28 values were zero) and 4 values were flagged as possible outliers.

Table 14. Mean of each item adjusted for the farm, with 95% confidence interval and comparison with the check Hybrid 04. Items that share a letter are not significantly different (p = 0.05).

Option	N	Mean	Lower 95%	Upper 95%	% of check	p vs check	Group
Hybrid 03	25	11.35	10.44	12.26	159.86	< 0.001	d
Hybrid 06	21	10.11	9.14	11.07	142.34	< 0.001	c
Hybrid 02	21	9.93	8.96	10.91	139.91	< 0.001	c
Hybrid 08	28	9.68	8.80	10.55	136.30	< 0.001	bc
Hybrid 05	34	9.19	8.36	10.02	129.40	< 0.001	bc
Hybrid 01	32	8.86	8.02	9.70	124.79	< 0.001	b
Hybrid 04	32	7.10	6.26	7.94	100.00		a
Hybrid 07	21	6.42	5.46	7.38	90.45	0.173	a

Broad-sense heritability is 0.96. Values close to 1 mean that most of the variation between plots comes from differences between the items, values close to 0 that it comes from the environment and measurement error.

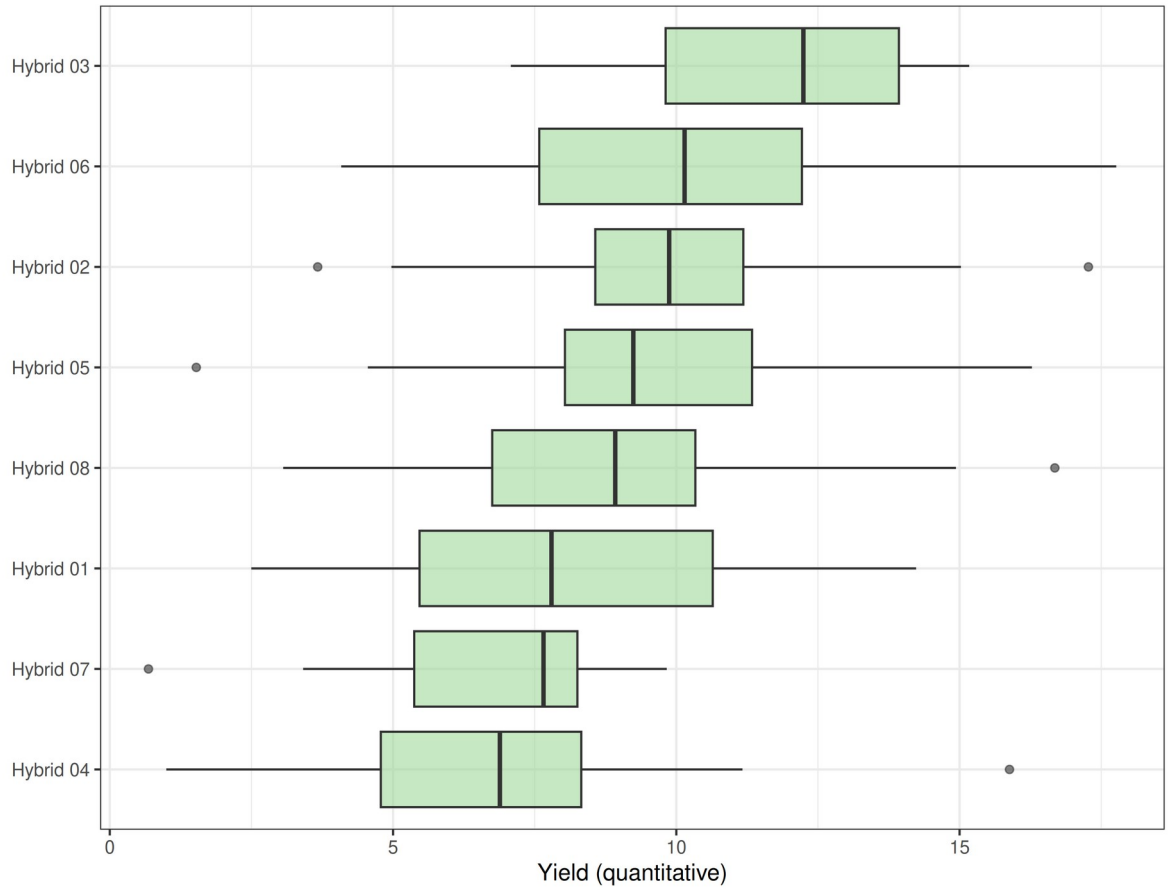


Figure 18. Distribution of Yield (quantitative) - Agronomic performance by option, possible outliers removed.

Evidence for variety release

Tables 15 and 16 bring together the evidence for each item against the check Hybrid 04, in a form that can support a variety release application. The probability of beating the check and its p-value come from the Plackett-Luce model for ‘Overall Preference’. The favourability score is described in the section on performance. The traits where each item was better or worse than the check are those with $p < 0.05$. When yield was measured, the mean and its percentage of the check come from the mixed model.

Table 15. Summary of each option against the check Hybrid 04.

Option	Times tested	Beats check	p-value	Favourability	Yield (mean)	Yield (% of check)
Hybrid 03	40	79%	0.00164	46.0	11.3	159.9
Hybrid 06	40	70%	0.0542	24.5	10.1	142.3
Hybrid 08	52	64%	0.136	8.8	9.7	136.3
Hybrid 02	38	62%	0.271	3.3	9.9	139.9
Hybrid 05	59	56%	0.551	-8.9	9.2	129.4
Hybrid 01	45	54%	0.653	-12.7	8.9	124.8
Hybrid 04	49			-23.5	7.1	100.0
Hybrid 07	40	49%	0.943	-23.0	6.4	90.4

Table 16. Traits in which each option was better or worse than the check Hybrid 04 ($p < 0.05$).

Option	Better than check in	Worse than check in
Hybrid 03	Germination; Diseases resistance; Drought tolerance; Yield; Grain size; Grain colour; Taste; Overall preference	
Hybrid 06	Germination; Diseases resistance; Drought tolerance; Fall armyworm resistance; Yield; Grain size; Grain colour	
Hybrid 08	Germination; Diseases resistance; Drought tolerance; Fall armyworm resistance; Grain colour; Taste; Marketability	
Hybrid 02	Germination; Diseases resistance; Drought tolerance; Fall armyworm resistance; Taste	
Hybrid 05	Germination	
Hybrid 01	Diseases resistance; Grain colour; Taste	
Hybrid 04		
Hybrid 07		Drought tolerance

The full comparison, trait by trait and for every check, is in the files ‘variety-release-summary.csv’ and ‘variety-release-by-trait.csv’ in the extra outputs.

What participants said in their own words

Besides ranking the options, participants answered open questions in their own words, by text or voice. Sikia Analytics read 84 answers and found 254 mentions of 9 different topics, such as a trait, a pest or a quality, each with the feeling expressed about it. The labels and sentiments are reported as Sikia returned them, with their counts, so readers can weigh them.

Figure 19 shows the 9 most mentioned topics. Bars to the right count positive mentions, bars to the left negative ones, and grey bars neutral ones. The number at the end of each bar is the total.

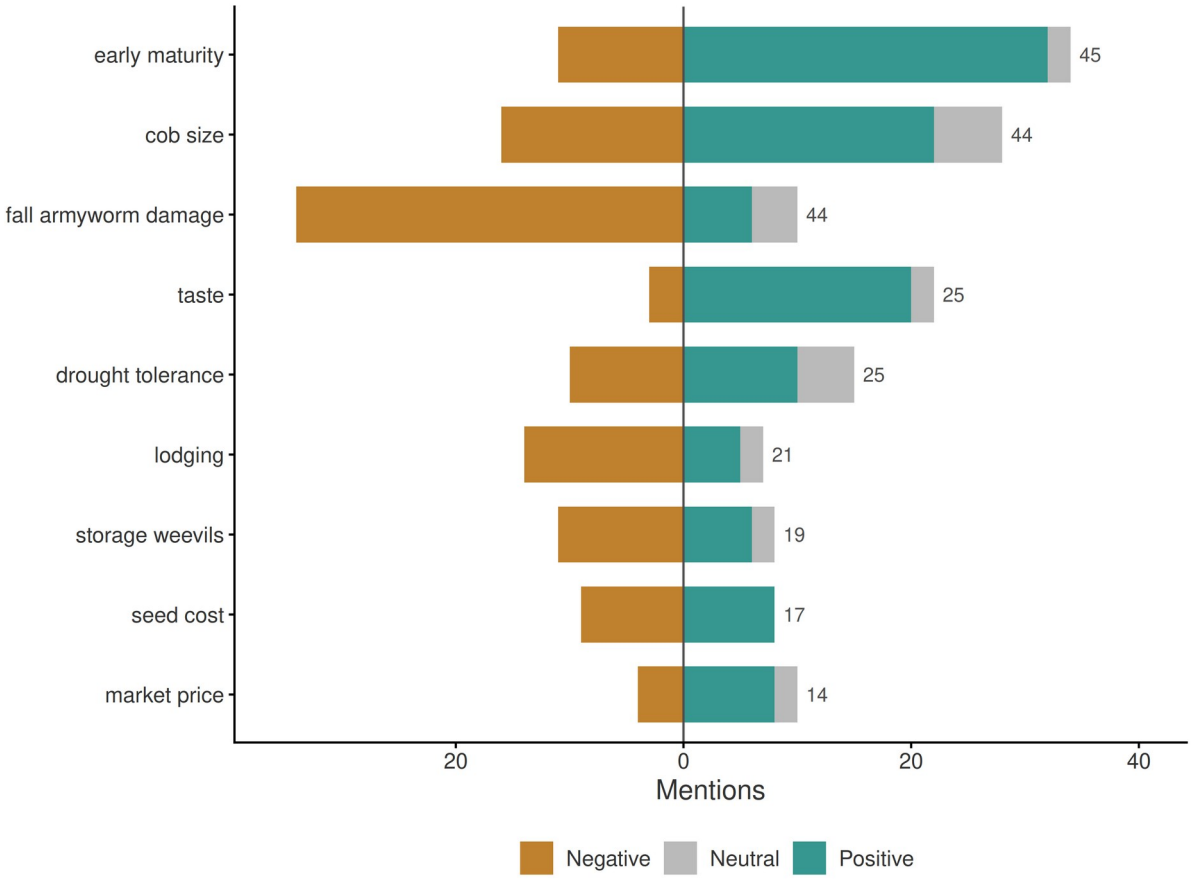


Figure 19. Topics mentioned by participants in their open answers, and the sentiment of each mention, as read by Sikia Analytics.

Sikia linked part of the mentions to the items they were about. Figure 20 shows, for each items and topic, the number of mentions and whether they were mostly positive (teal) or negative (brown).

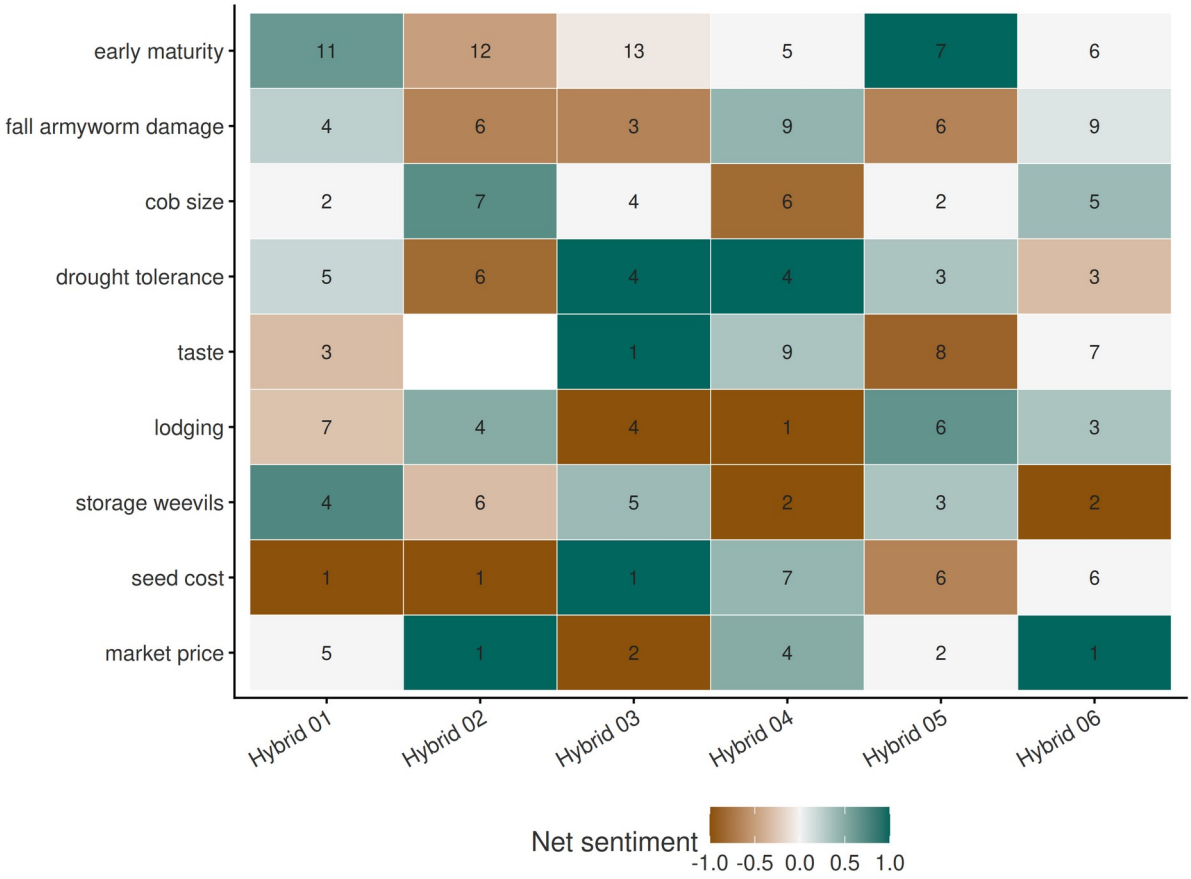


Figure 20. Mentions by Item and topic. Numbers are mentions, colour is the net sentiment, from all negative (brown) to all positive (teal).

In their words

One positive and one negative quote for the most mentioned topics, chosen by Sikia's confidence in the reading. They are the words of individual participants, not a summary.

Table 17. Quotes from the open answers, by topic and sentiment.

Topic	Sentiment	Quote
early maturity	Positive	ready before the rains stopped with Hybrid 01
early maturity	Negative	still green when the others were dry with Hybrid 01
cob size	Positive	cobs filled to the tip with Hybrid 05
cob size	Negative	small cobs with Hybrid 01
fall armyworm damage	Positive	few worms on this one with Hybrid 01
fall armyworm damage	Negative	armyworm ate the leaves with Hybrid 01
taste	Positive	soft and tasty with Hybrid 02
taste	Negative	hard to chew with Hybrid 04
drought tolerance	Positive	survived the dry spell in February with Hybrid 02
drought tolerance	Negative	wilted when the rains failed with Hybrid 01

Topics across all answers

Sikia grouped similar mentions into 7 topics in 3 broader categories (Table 18). Topics show what participants kept returning to, including things the trial did not ask about.

Table 18. Topics found by Sikia across all open answers, with the number of mentions and representative phrases.

Category	Topic	Mentions	Representative phrases
Growing season	Early maturity and escape from drought	70	ready before the rains stopped · kept green with little rain · harvested earliest of the three
Pests in field and store	Fall armyworm and spraying	44	armyworm ate the leaves · had to spray twice
Harvest, food and money	Cob size and filling	44	big cobs with full rows · many empty tips
Harvest, food and money	Seed cost and market	31	sold at a good price · seed too expensive for next season
Harvest, food and money	Eating quality	25	sweet when roasted · children liked the ugali from it
Growing season	Standing in wind	21	stood well in the wind · fell over after the storm
Pests in field and store	Weevils in storage	19	stored for months without weevils · weevils came within weeks

Sikia’s reading of free text is automatic and can miss or misread an answer. Counts of mentions are not counts of participants, as one answer can carry several mentions. The results were produced by Sikia Analytics on 2026-10-20 and are shared through ClimMob. The mentions table is in extra-outputs/sikia-mentions.csv.

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Appendix

Monitoring tables

Table 19. Number of participants who answered each data collection moment, percentage of those registered and number lost since the previous moment.

Group	Data collection moment	Day	N	% of registered	Lost since previous
All participants	Registration	0	117	100.0	
All participants	Vegetative	30	111	94.9	6
All participants	Reproductive	70	106	90.6	5
All participants	Agronomic performance	85	82	70.1	24
Man	Registration	0	99	100.0	
Man	Vegetative	30	95	96.0	4
Man	Reproductive	70	89	89.9	6
Man	Agronomic performance	85	68	68.7	21
Prefer Not To Respond	Registration	0	1	100.0	
Prefer Not To Respond	Vegetative	30	1	100.0	0
Prefer Not To Respond	Reproductive	70	1	100.0	0
Prefer Not To Respond	Agronomic performance	85	1	100.0	0
Woman	Registration	0	17	100.0	
Woman	Vegetative	30	15	88.2	2
Woman	Reproductive	70	16	94.1	-1
Woman	Agronomic performance	85	13	76.5	3

Table 20. Forms submitted by data collector and data collection moment.

Data collection moment	Data collector	Forms
Registration	enumerator1	52
Registration	enumerator2	65
Vegetative	enumerator1	60
Vegetative	enumerator2	51
Reproductive	enumerator1	55
Reproductive	enumerator2	51
Agronomic performance	enumerator1	48
Agronomic performance	enumerator2	34

Traits analysed

Table 21. Traits analysed and number of valid answers.

Trait	Data collection moment	Question asked	Valid answers
Germination	Vegetative	Which option germinated more plants?, Which option germinated less plants?	101

Trait	Data collection moment	Question asked	Valid answers
Diseases resistance	Vegetative	Which option is more resistant to diseases?, Which option is more susceptible to diseases?	61
Drought tolerance	Vegetative	Which option is more tolerant to drought?, Which option is more susceptible to drought?	62
Fall armyworm resistance	Vegetative	Which option is more resistant to fall armyworm?, Which option is more susceptible to fall armyworm?	82
Diseases resistance	Reproductive	Which option is more resistant to diseases?, Which option is more susceptible to diseases?	55
Drought tolerance	Reproductive	Which option is more tolerant to drought?, Which option is more susceptible to drought?	46
Fall armyworm resistance	Reproductive	Which option is more resistant to fall armyworm?, Which option is more susceptible to fall armyworm?	81
Maturity	Agronomic Performance	Which option has the earlier maturity?, Which option has the later maturity?	78
Yield	Agronomic Performance	Which option has the higher grain yield?, Which option has the lower grain yield?	79
Grain size	Agronomic Performance	Which option has the best grain/seed size for the target market?, Which option has the worst grain/seed size for the target market?	77
Grain colour	Agronomic Performance	Which option has the best grain/seed colour for the target market?, Which option has the worst grain/seed colour for the target market?	78
Taste	Agronomic Performance	Which option has the best taste?, Which option has the worst taste?	77
Marketability	Agronomic Performance	Which option is easier to sell?, Which option is more difficult to sell?	72
Overall preference	Agronomic Performance	Overall, which option was the best?, Overall, which option was the worst?	75

Favourability by trait

Table 22. Favourability of the items, by trait.

Trait	Option	N	Top (%)	Bottom (%)	Favourability
Germination [Vegetative]	Hybrid 08	82	46.4	18.4	28.1
	Hybrid 02	66	42.6	22.1	20.5
	Hybrid 06	66	42.0	21.9	20.1
	Hybrid 03	72	39.2	25.1	14.1
	Hybrid 05	96	35.3	28.7	6.6
	Hybrid 01	74	26.1	39.9	-13.8

Trait	Option	N	Top (%)	Bottom (%)	Favourability
	Hybrid 07	68	18.3	53.9	-35.6
	Hybrid 04	82	17.3	56.2	-38.9
Diseases resistance [Vegetative]	Hybrid 03	36	57.4	11.0	46.4
	Hybrid 08	44	53.8	12.9	41.0
	Hybrid 02	42	45.7	19.8	25.8
	Hybrid 06	40	40.6	22.4	18.2
	Hybrid 01	48	33.6	25.5	8.0
	Hybrid 05	56	25.9	33.6	-7.7
	Hybrid 04	60	13.9	57.7	-43.8
	Hybrid 07	40	8.2	73.4	-65.2
Drought tolerance [Vegetative]	Hybrid 03	40	55.6	11.9	43.7
	Hybrid 06	52	44.0	19.4	24.6
	Hybrid 08	52	39.8	24.2	15.6
	Hybrid 02	32	36.4	28.1	8.3
	Hybrid 01	48	32.0	30.7	1.3
	Hybrid 05	62	28.0	36.1	-8.1
	Hybrid 04	42	16.1	58.1	-42.0
	Hybrid 07	44	16.0	59.2	-43.2
Fall armyworm resistance [Vegetative]	Hybrid 06	48	48.8	16.8	32.1
	Hybrid 03	54	42.3	22.2	20.1
	Hybrid 05	82	37.3	27.3	10.1
	Hybrid 01	60	35.7	28.1	7.6
	Hybrid 02	48	34.0	31.6	2.4
	Hybrid 08	80	29.6	35.9	-6.3
	Hybrid 04	60	27.7	39.5	-11.9
	Hybrid 07	60	15.1	61.9	-46.8
Diseases resistance [Reproductive]	Hybrid 03	36	48.2	17.5	30.7
	Hybrid 06	24	45.6	19.9	25.7
	Hybrid 02	32	38.0	26.2	11.8
	Hybrid 08	58	37.8	27.1	10.7
	Hybrid 01	46	27.9	39.4	-11.5
	Hybrid 07	40	26.9	40.9	-14.0
	Hybrid 04	46	25.9	42.2	-16.2
	Hybrid 05	48	25.3	43.6	-18.3
Drought tolerance [Reproductive]	Hybrid 03	32	50.8	14.9	35.9
	Hybrid 08	42	51.0	15.8	35.2
	Hybrid 04	30	39.3	23.9	15.4
	Hybrid 02	36	32.8	29.4	3.4
	Hybrid 01	38	33.3	31.2	2.1
	Hybrid 06	22	28.6	38.4	-9.8
	Hybrid 05	46	16.3	54.3	-38.0

Trait	Option	N	Top (%)	Bottom (%)	Favourability
	Hybrid 07	30	14.2	58.5	-44.3
Fall armyworm resistance [Reproductive]	Hybrid 02	54	43.7	21.2	22.5
	Hybrid 08	66	42.7	21.6	21.1
	Hybrid 06	60	42.5	22.2	20.2
	Hybrid 05	78	33.3	31.0	2.3
	Hybrid 03	60	33.3	31.9	1.3
	Hybrid 01	60	31.3	33.5	-2.2
	Hybrid 04	54	21.6	48.9	-27.3
	Hybrid 07	54	15.4	61.2	-45.9
Maturity [Agronomic Performance]	Hybrid 03	56	48.5	17.2	31.3
	Hybrid 02	44	47.6	17.6	30.1
	Hybrid 06	48	47.1	18.6	28.5
	Hybrid 08	66	33.5	30.6	2.9
	Hybrid 04	64	29.6	35.7	-6.1
	Hybrid 05	72	27.9	38.8	-10.9
	Hybrid 01	64	23.7	43.4	-19.6
	Hybrid 07	54	16.6	57.4	-40.8
Yield [Agronomic Performance]	Hybrid 03	56	50.0	16.1	33.9
	Hybrid 06	52	49.7	15.9	33.7
	Hybrid 08	66	37.1	26.5	10.6
	Hybrid 02	40	36.5	28.5	8.1
	Hybrid 05	76	29.6	35.4	-5.8
	Hybrid 01	62	28.6	34.8	-6.1
	Hybrid 04	70	26.3	41.1	-14.8
	Hybrid 07	52	12.4	66.6	-54.2
Grain size [Agronomic Performance]	Hybrid 03	56	51.2	14.9	36.3
	Hybrid 06	50	43.1	21.0	22.1
	Hybrid 02	44	40.9	23.2	17.8
	Hybrid 08	62	39.2	24.5	14.7
	Hybrid 01	60	30.8	33.2	-2.5
	Hybrid 05	74	27.5	39.0	-11.5
	Hybrid 04	64	21.8	48.5	-26.7
	Hybrid 07	52	16.8	57.6	-40.8
Grain colour [Agronomic Performance]	Hybrid 03	56	47.1	17.7	29.3
	Hybrid 06	50	43.4	20.3	23.1
	Hybrid 08	66	42.8	21.2	21.6
	Hybrid 01	62	36.3	26.5	9.8
	Hybrid 02	42	35.3	29.2	6.0
	Hybrid 05	72	30.8	33.1	-2.3
	Hybrid 04	68	17.9	55.7	-37.7
	Hybrid 07	52	15.4	60.6	-45.2

Trait	Option	N	Top (%)	Bottom (%)	Favourability
Taste [Agronomic Performance]	Hybrid 03	54	58.2	10.5	47.7
	Hybrid 08	70	40.9	22.7	18.2
	Hybrid 01	58	36.3	26.2	10.1
	Hybrid 02	40	34.9	28.4	6.5
	Hybrid 06	50	31.7	31.9	-0.3
	Hybrid 05	74	30.8	33.1	-2.3
	Hybrid 07	50	19.0	53.6	-34.6
	Hybrid 04	66	16.3	58.4	-42.1
Marketability [Agronomic Performance]	Hybrid 08	60	48.6	17.0	31.6
	Hybrid 06	44	40.3	23.7	16.7
	Hybrid 05	64	40.0	24.4	15.6
	Hybrid 02	42	37.4	27.5	9.8
	Hybrid 03	54	29.8	36.6	-6.8
	Hybrid 01	54	28.5	37.0	-8.5
	Hybrid 04	62	22.1	47.9	-25.8
	Hybrid 07	52	20.4	51.4	-31.1
Overall preference [Agronomic Performance]	Hybrid 03	54	57.3	11.3	46.0
	Hybrid 06	42	44.6	20.1	24.5
	Hybrid 08	62	36.6	27.7	8.8
	Hybrid 02	42	33.8	30.5	3.3
	Hybrid 05	70	28.6	37.5	-8.9
	Hybrid 01	58	26.5	39.2	-12.7
	Hybrid 07	54	23.0	46.0	-23.0
	Hybrid 04	68	23.1	46.5	-23.5

Kendall correlation

Table 23. Kendall correlation between 'Overall Preference' and the other traits. A p-value below 0.05 indicates a significant correlation.

Trait	Kendall tau	Lower 95%	Upper 95%	Z value	Pr(>|z|)
Drought tolerance [Vegetative]	0.136	-0.005	0.254	0.847	1.98e-01
Maturity [Agronomic performance]	0.125	-0.008	0.247	0.848	1.98e-01
Germination [Vegetative]	0.120	0.004	0.247	0.801	2.11e-01
Diseases resistance [Reproductive]	0.120	0.005	0.241	0.743	2.29e-01
Taste [Agronomic performance]	0.115	0.012	0.214	0.775	2.19e-01
Diseases resistance [Vegetative]	0.106	-0.020	0.212	0.651	2.57e-01
Grain colour [Agronomic performance]	0.095	-0.061	0.200	0.642	2.60e-01
Marketability [Agronomic performance]	0.051	-0.069	0.158	0.341	3.66e-01
Drought tolerance [Reproductive]	0.033	-0.083	0.145	0.202	4.20e-01
Fall armyworm resistance [Reproductive]	0.028	-0.081	0.131	0.179	4.29e-01
Yield [Agronomic performance]	-0.023	-0.123	0.090	-0.153	5.61e-01

Trait	Kendall tau	Lower 95%	Upper 95%	Z value	Pr(>|z|)
Grain size [Agronomic performance]	-0.023	-0.116	0.094	-0.155	5.62e-01
Fall armyworm resistance [Vegetative]	-0.053	-0.166	0.064	-0.345	6.35e-01

Possible outliers in measured traits

Table 24. Possible outliers in the measured traits. If any of these values is a mistake, correct it in ClimMob and request a new report.

id	technology	trait	data_collection	value	enumerator
4	Hybrid 05	Plant population	Reproductive	66.000	enumerator 1
4	Hybrid 07	Disease severity	Reproductive	59.000	enumerator 1
7	Hybrid 01	Plant population	Reproductive	28.000	enumerator 1
18	Hybrid 07	Number of cobs	Agronomic performance	15.000	enumerator 2
34	Hybrid 06	Yield (quantitative)	Agronomic performance	72.988	enumerator 2
35	Hybrid 07	Disease severity	Vegetative	54.000	enumerator 1
36	Hybrid 08	Plant population	Reproductive	67.000	enumerator 2
41	Hybrid 06	Plant population	Vegetative	68.000	enumerator 1
50	Hybrid 05	Disease severity	Vegetative	53.000	enumerator 2
57	Hybrid 07	Disease severity	Vegetative	55.000	enumerator 1
62	Hybrid 04	Disease severity	Vegetative	56.000	enumerator 1
72	Hybrid 05	Number of cobs	Agronomic performance	13.000	enumerator 2
97	Hybrid 08	Yield (quantitative)	Agronomic performance	137.143	enumerator 1

Climate indices

Figure 21 and Figure 22 show the weekly temperature and rainfall indices at the trial plots, every plot as a grey line and the smoothed trend in green. Figure 23 places every plot in the space of total rainfall and maximum day temperature over the window, the climate envelope of the trial.

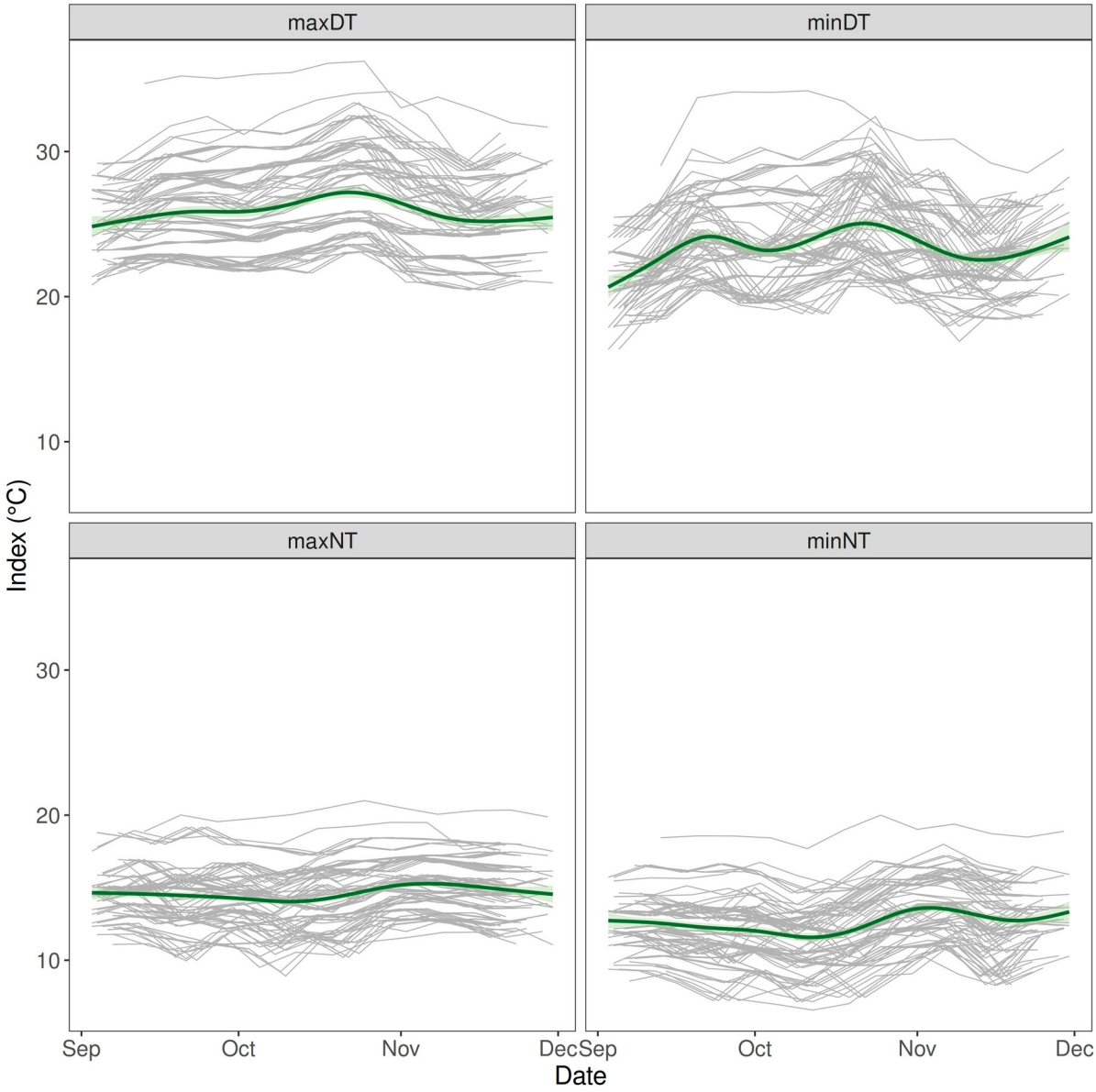


Figure 21. Weekly temperature indices from 2022-09-03 to 2022-12-08. Maximum day temperature (maxDT), minimum day temperature (minDT), maximum night temperature (maxNT) and minimum night temperature (minNT).

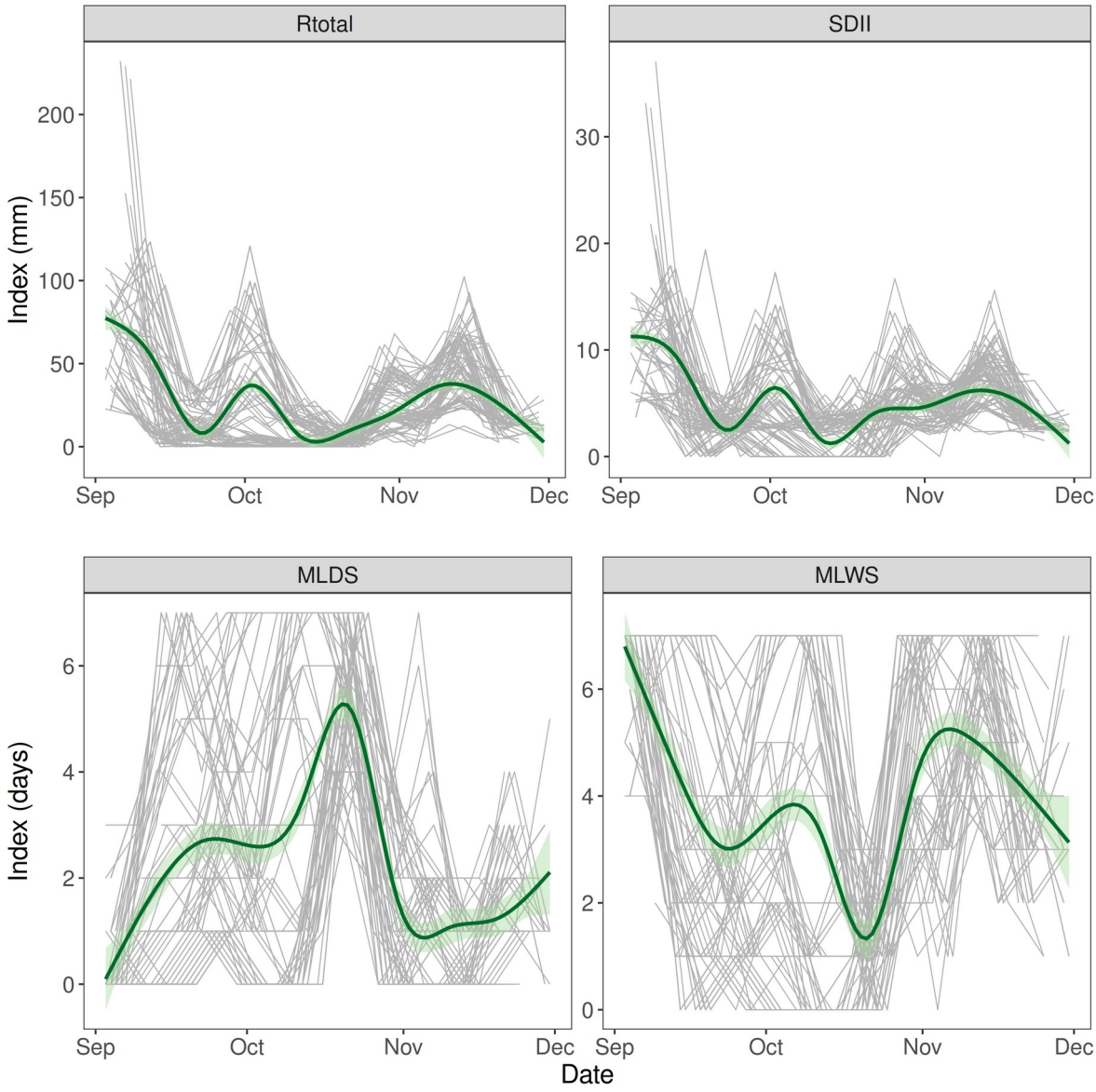


Figure 22. Weekly rainfall indices from 2022-09-03 to 2022-12-08. Total rainfall (Rtotal), daily intensity on wet days (SDII), longest dry spell with rain below 1 mm (MLDS) and longest wet spell with rain of 1 mm or more (MLWS).

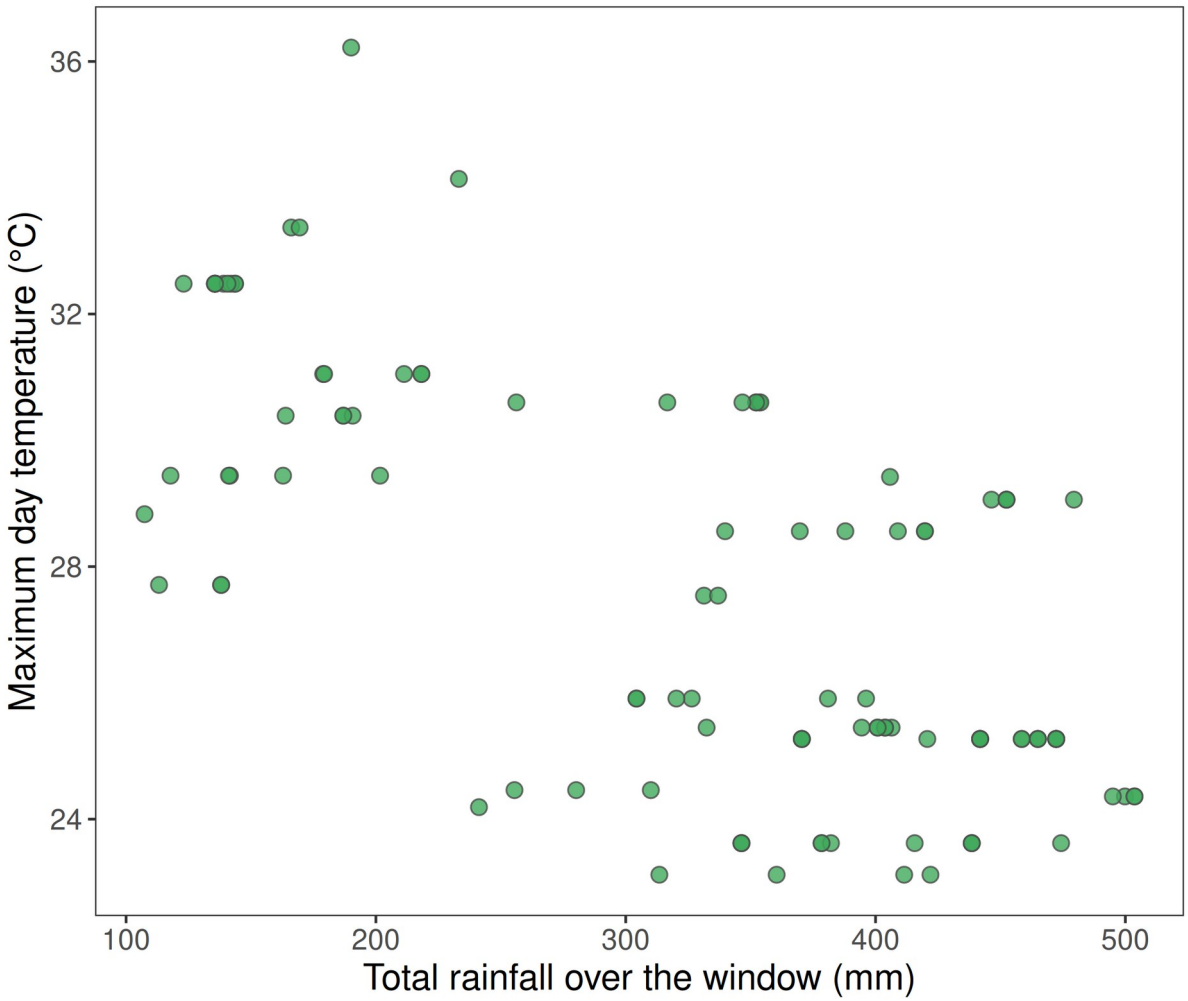


Figure 23. Climate envelope of the trial. Every trial plot by its total rainfall and maximum day temperature from 2022-09-03 to 2022-12-08.